

# Practice Test I (Version 2)

Name  
Date  
Course

I) Write your name, the date, and the course title (i.e. Trigonometry).

II) Show all work.

III) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.

IV) You will need a calculator for this test (be sure that it is in degree mode).

V) Each problem is worth four points for a total of 100 points.

① The measure of an angle in standard position is written below. Find the measures of three coterminal angles in standard position.

$$131^\circ$$

$$\theta_1$$

$$\theta_2$$

$$\theta_3$$

② If  $\cos \theta = 1$ , and  $0 \leq \theta \leq 360^\circ$ , find the values of  $\theta$ .

③ If  $\tan \theta = 0$ , find all values of  $\theta$ .

④ If  $\sec \theta = -1$ , and  $0 \leq \theta \leq 360^\circ$ , find the values of  $\theta$ .

⑤ Find the values of the expressions below with a calculator. Round to the nearest thousandth.

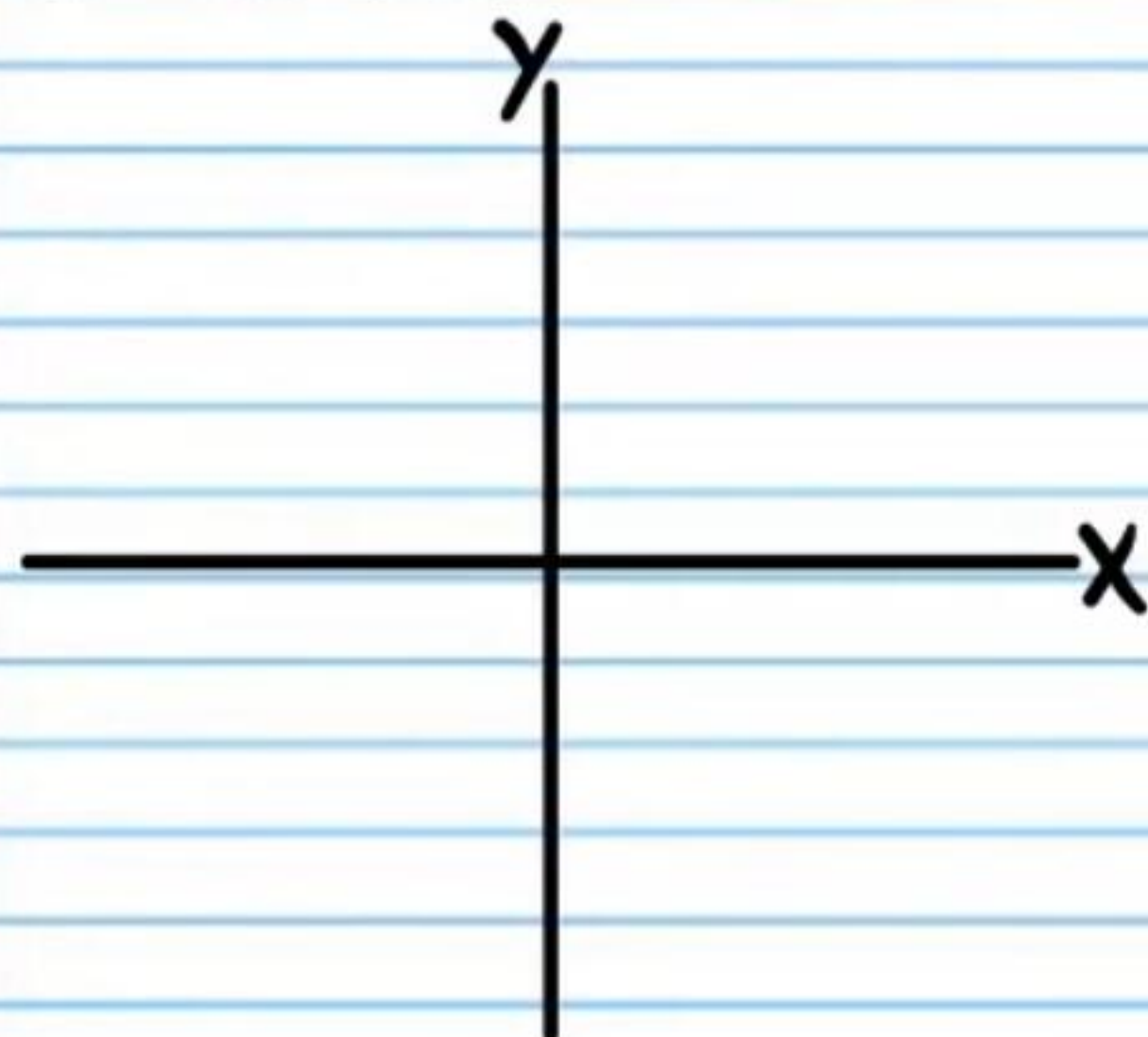
i)  $\cos 198^\circ$

ii)  $\tan 81^\circ$

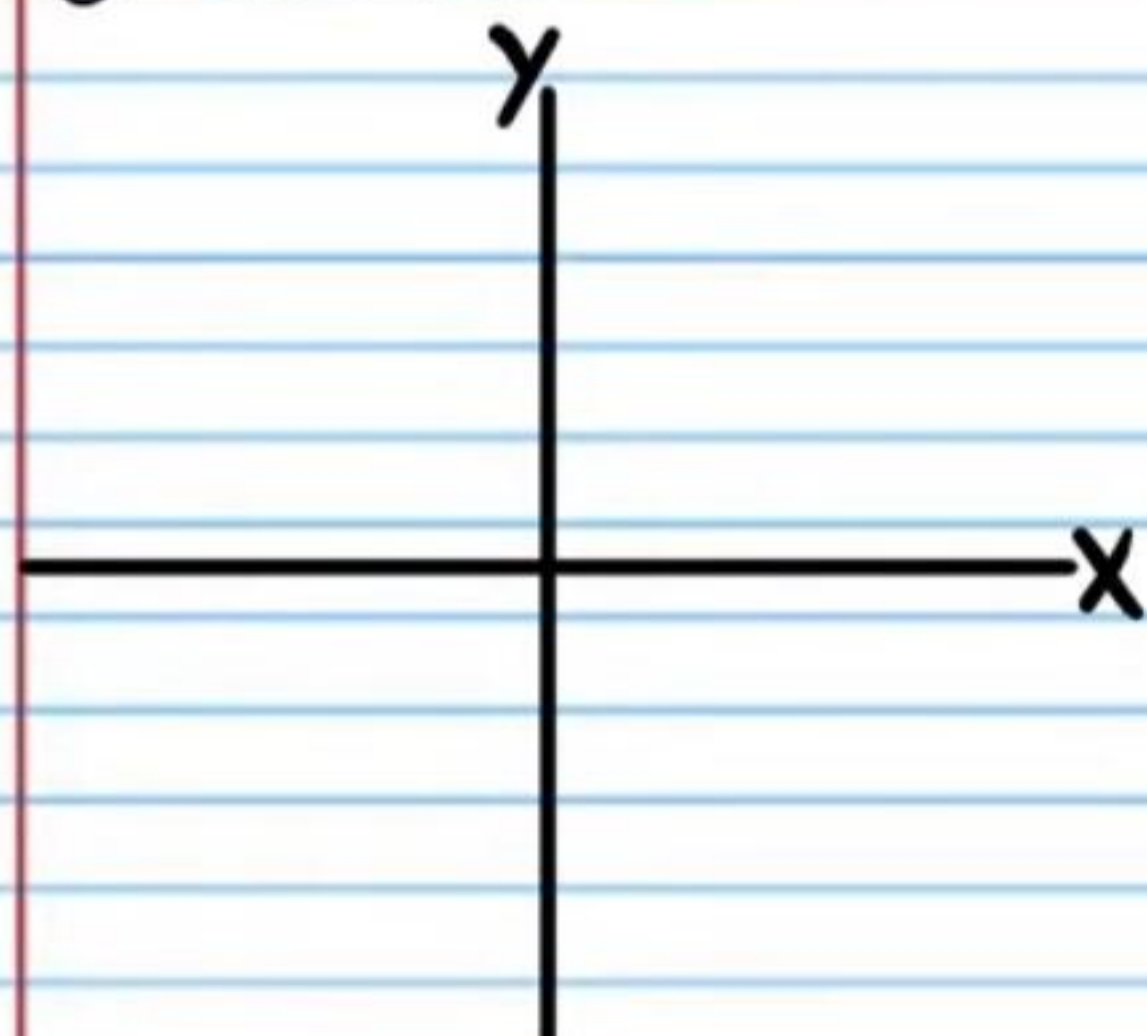


Find the values of the following expressions *without a calculator*. You must write exact answers. Draw each angle in standard position. Write the measure of each angle and the measure of its reference angle, along with the corresponding right triangle. *If a triangle is collapsed, then you cannot draw it. In this scenario, draw only the angle.*

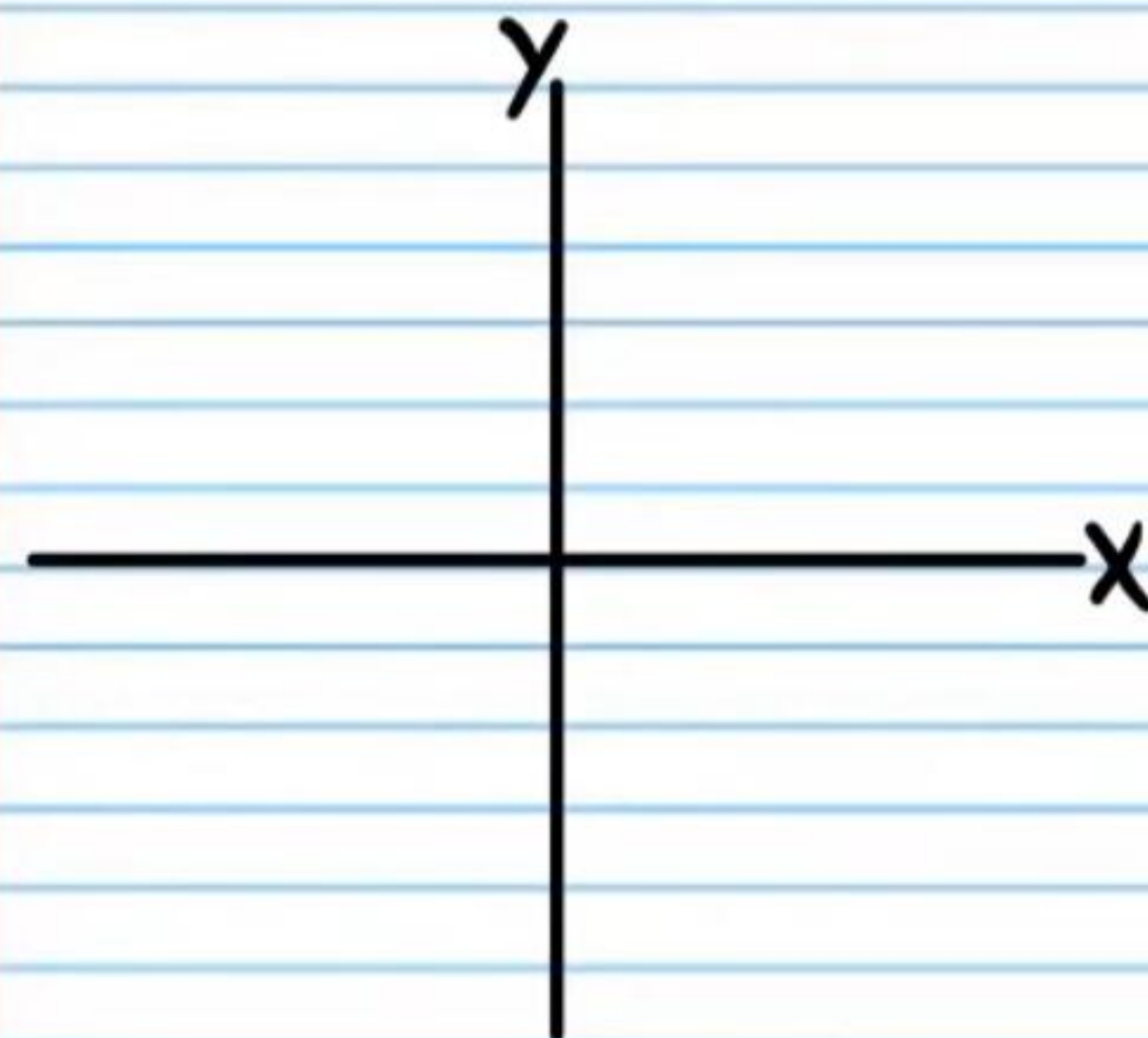
⑥  $\tan 300^\circ$



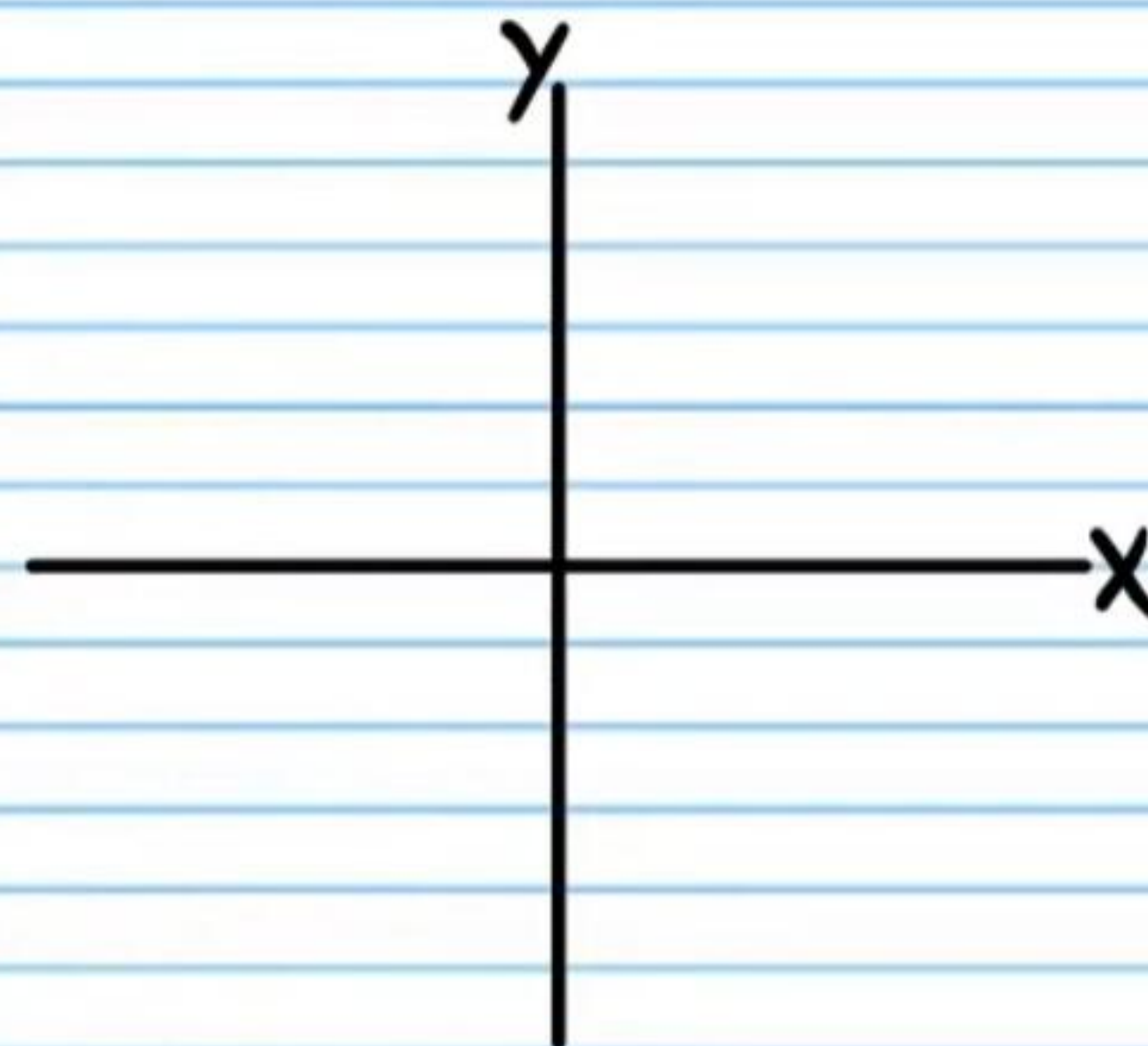
⑦  $\cot 135^\circ$



⑧  $\cos(-150^\circ)$

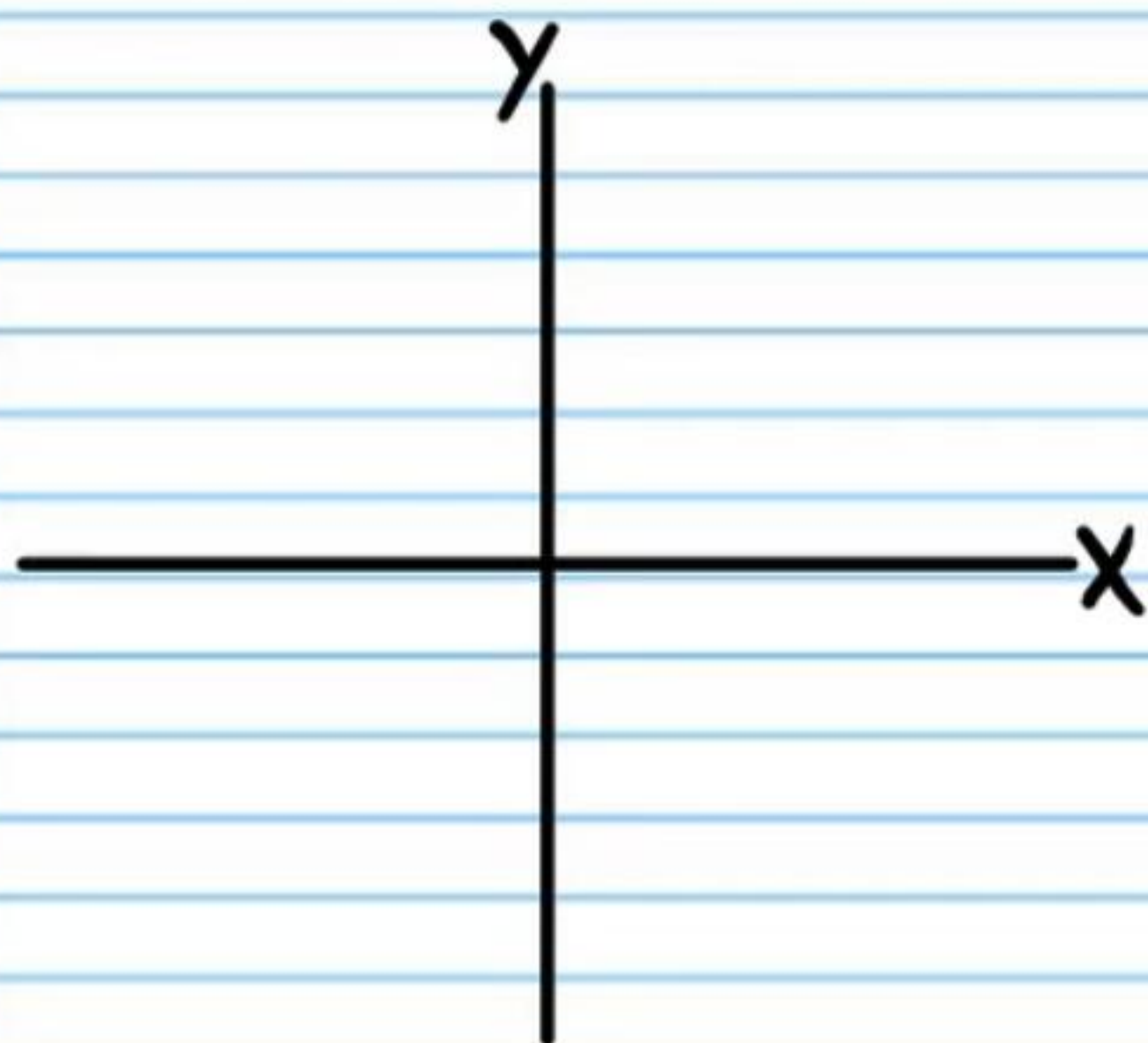


⑨  $\csc 0^\circ$

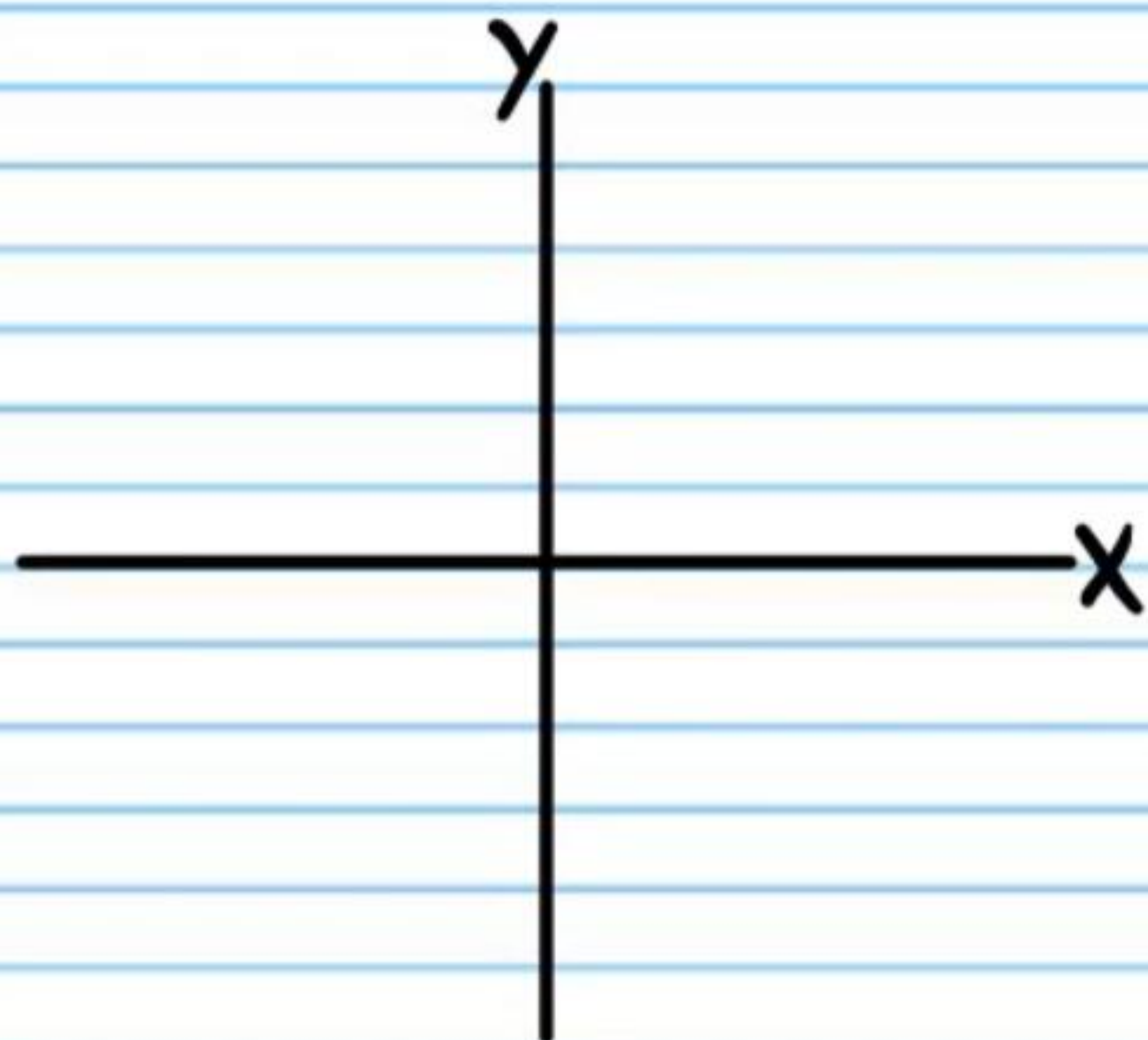




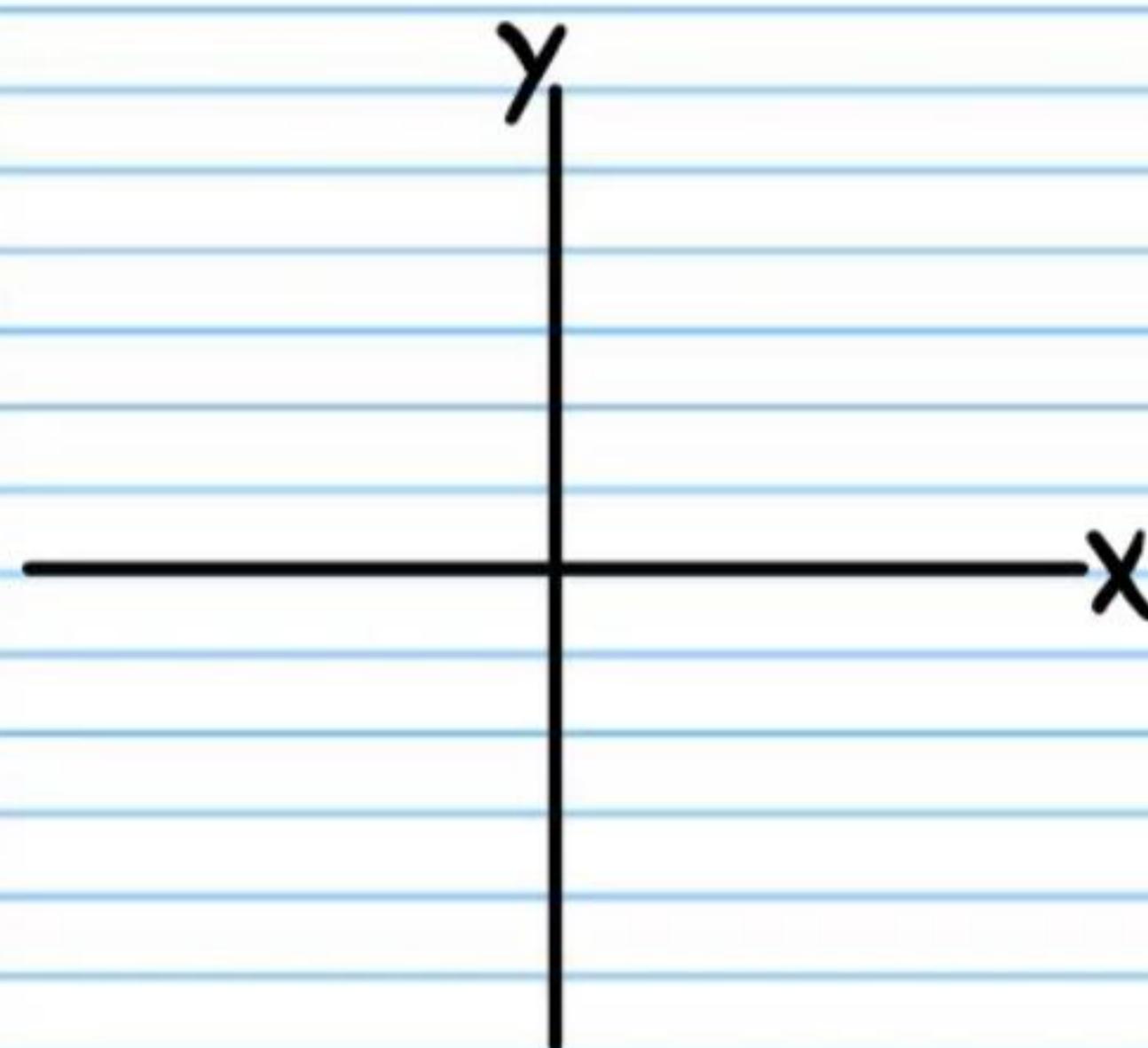
⑩  $\sin 90^\circ$



⑪  $\sec(-180^\circ)$



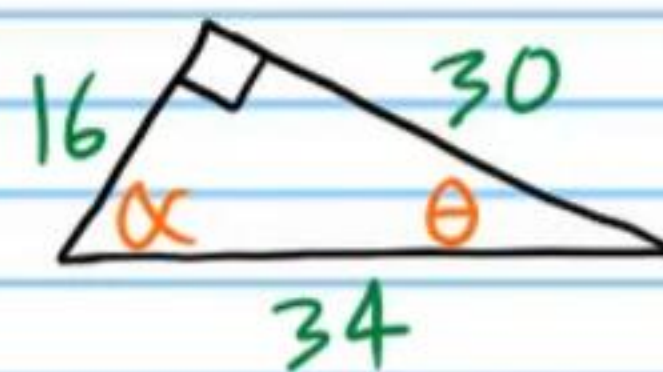
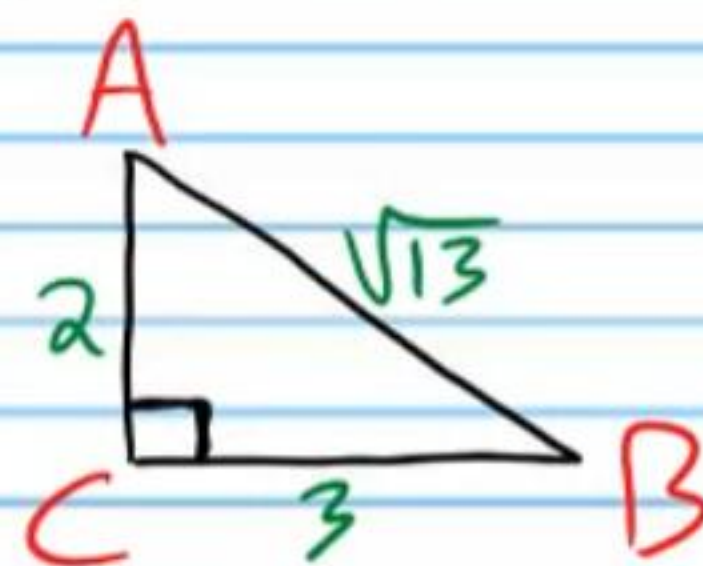
⑫  $\csc 60^\circ$



⑬ Use the triangles below to find the values of the expressions. *If necessary, rationalize denominators.*

i)  $\sin B$

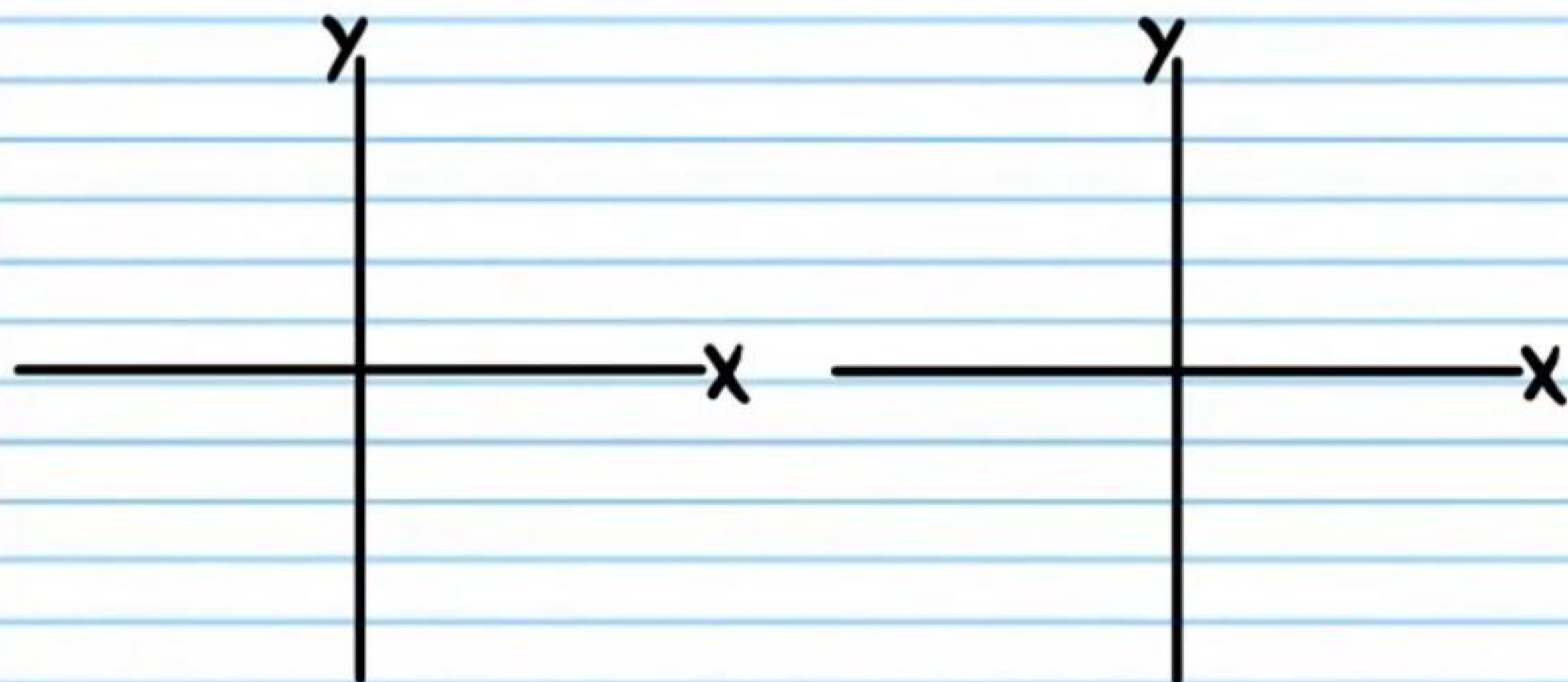
ii)  $\sec \alpha$



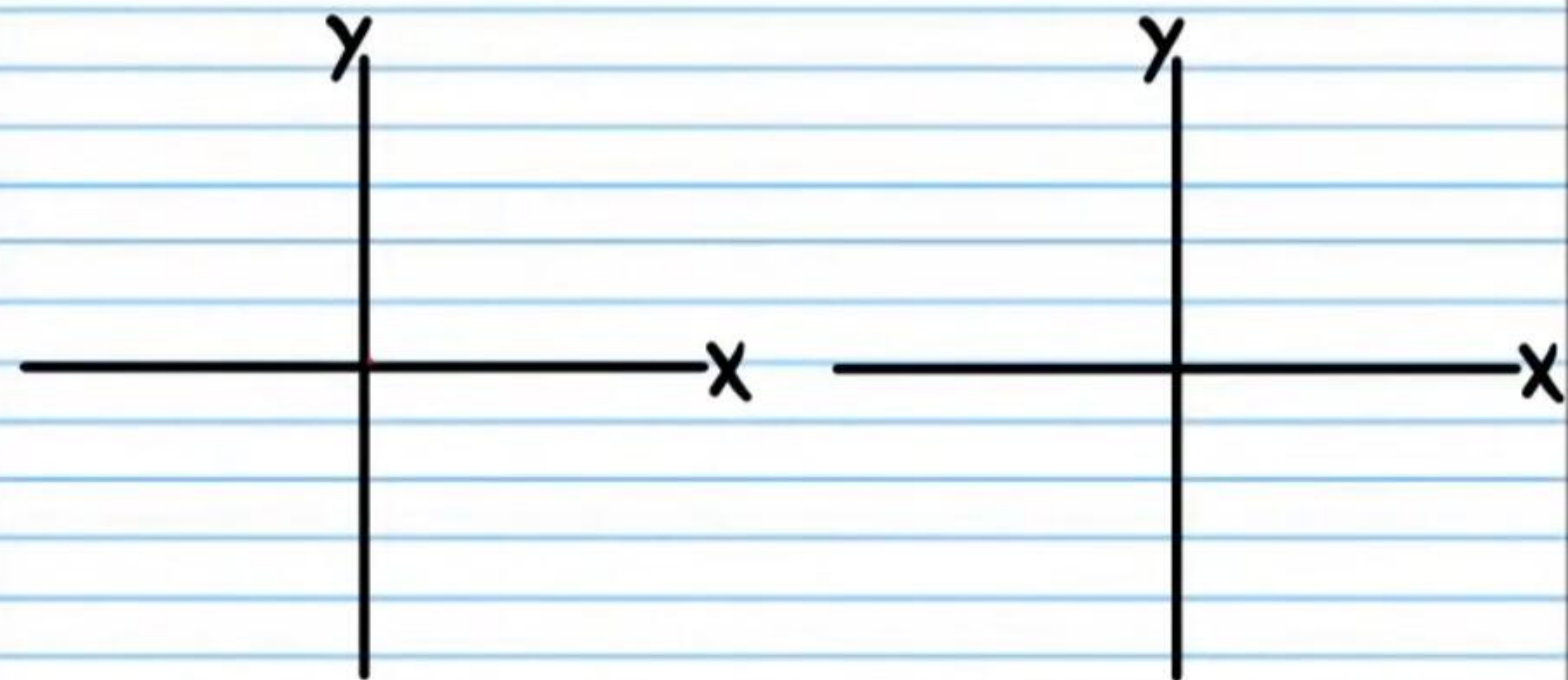


For problem #'s 14-18, draw the angles that are solutions to each equation in Standard position as well as their corresponding right triangles. Write all three interior angle measures of each triangle and the relative side lengths.  
*Do not use a calculator.*

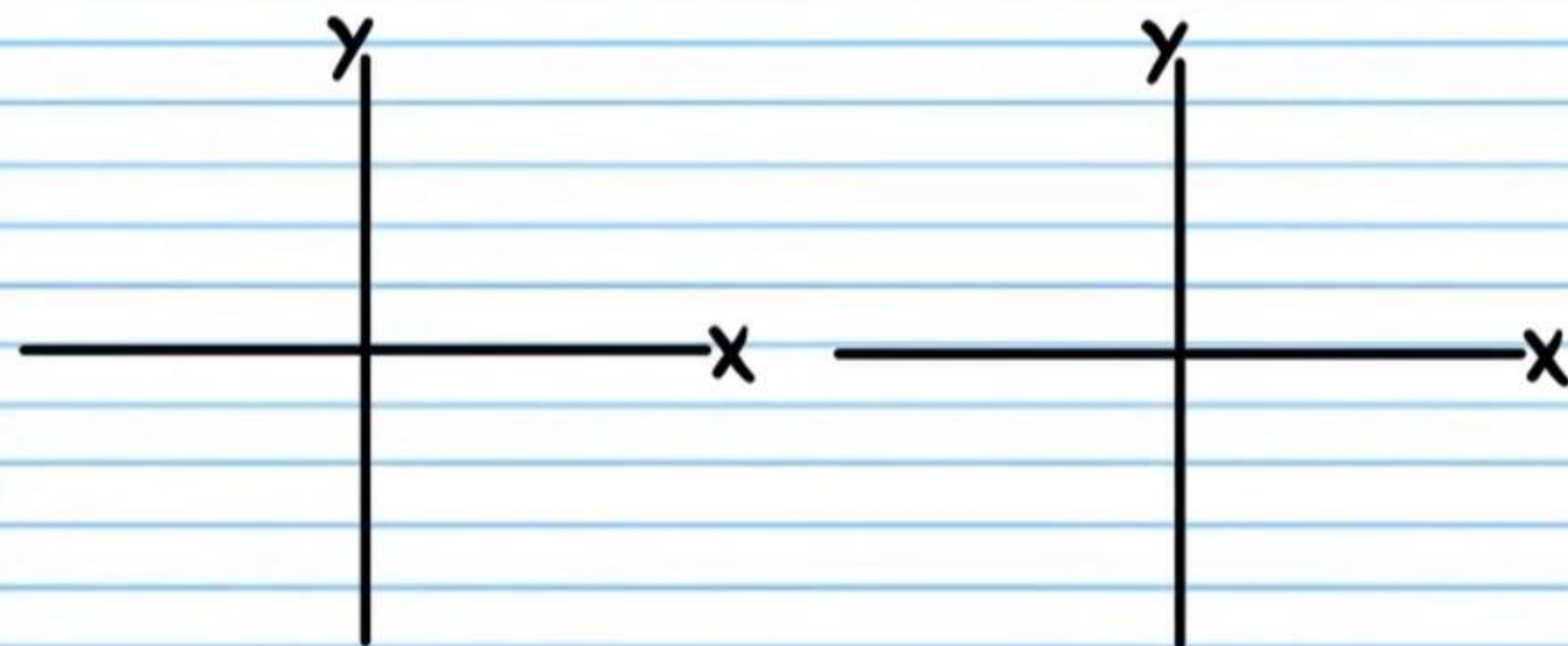
⑭ If  $\csc \theta = -2$ , and  $0^\circ < \theta \leq 360^\circ$ , Find the values of  $\theta$ .



⑮ If  $\cot \theta = -\frac{1}{\sqrt{3}}$ , and  $0^\circ < \theta \leq 360^\circ$ , Find the values of  $\theta$ .

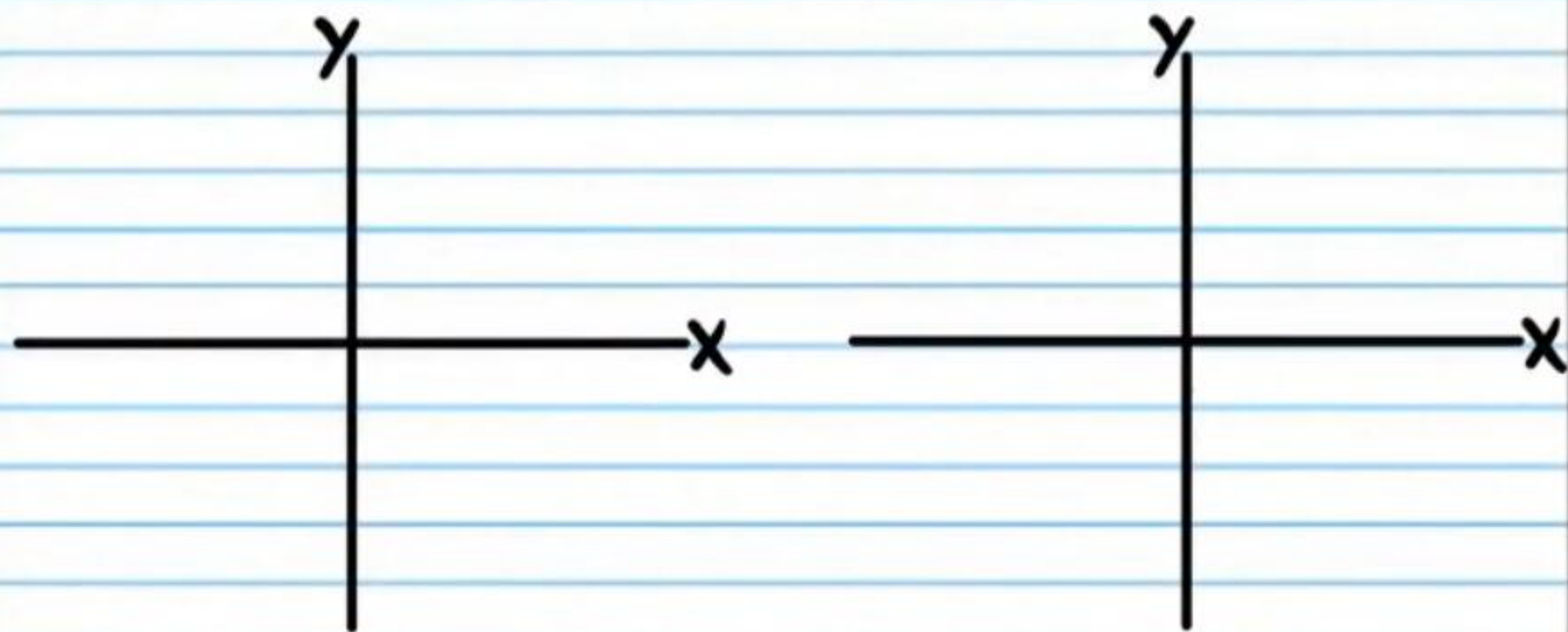


⑯ If  $\tan \theta = 1$ , and  $0^\circ < \theta \leq 360^\circ$ , Find the values of  $\theta$ .

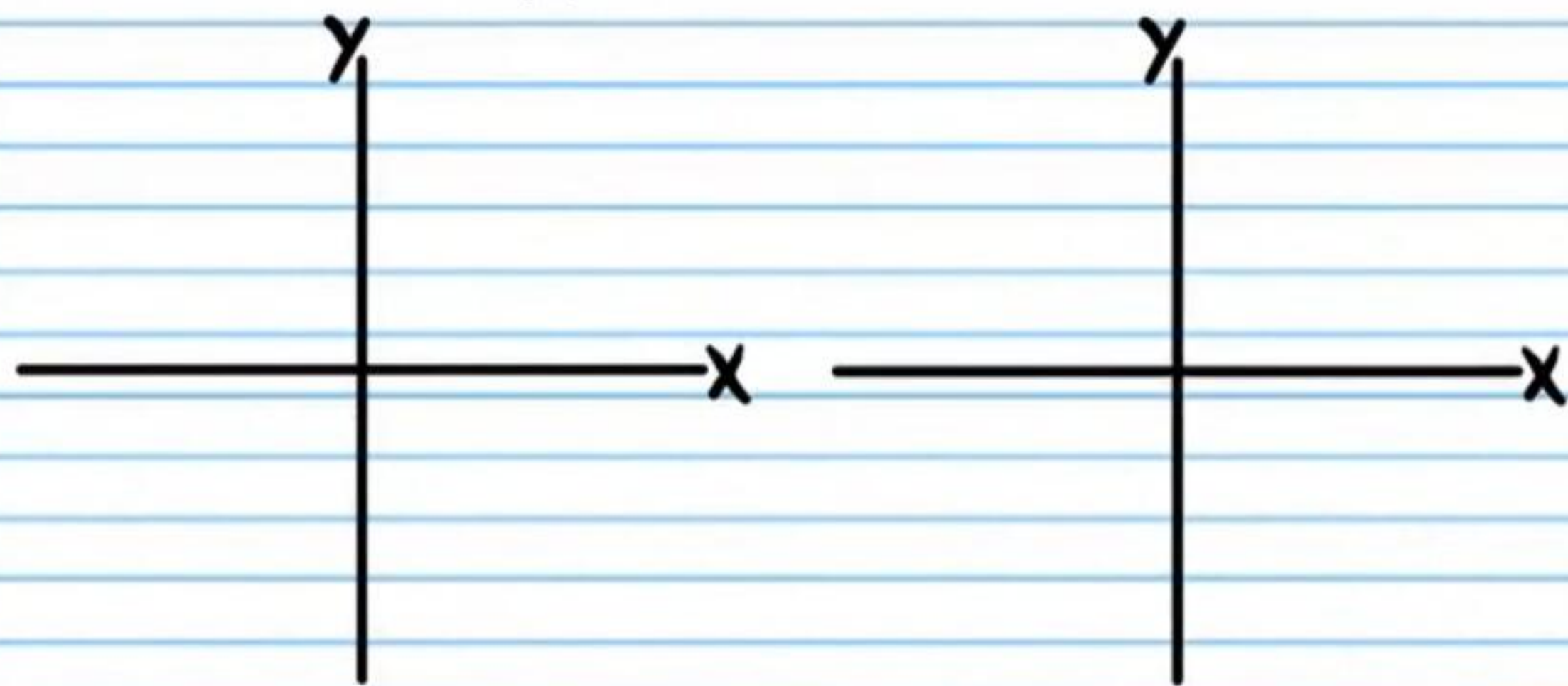




⑪ If  $\sin \theta = \frac{1}{2}$ , find all values of  $\theta$ .

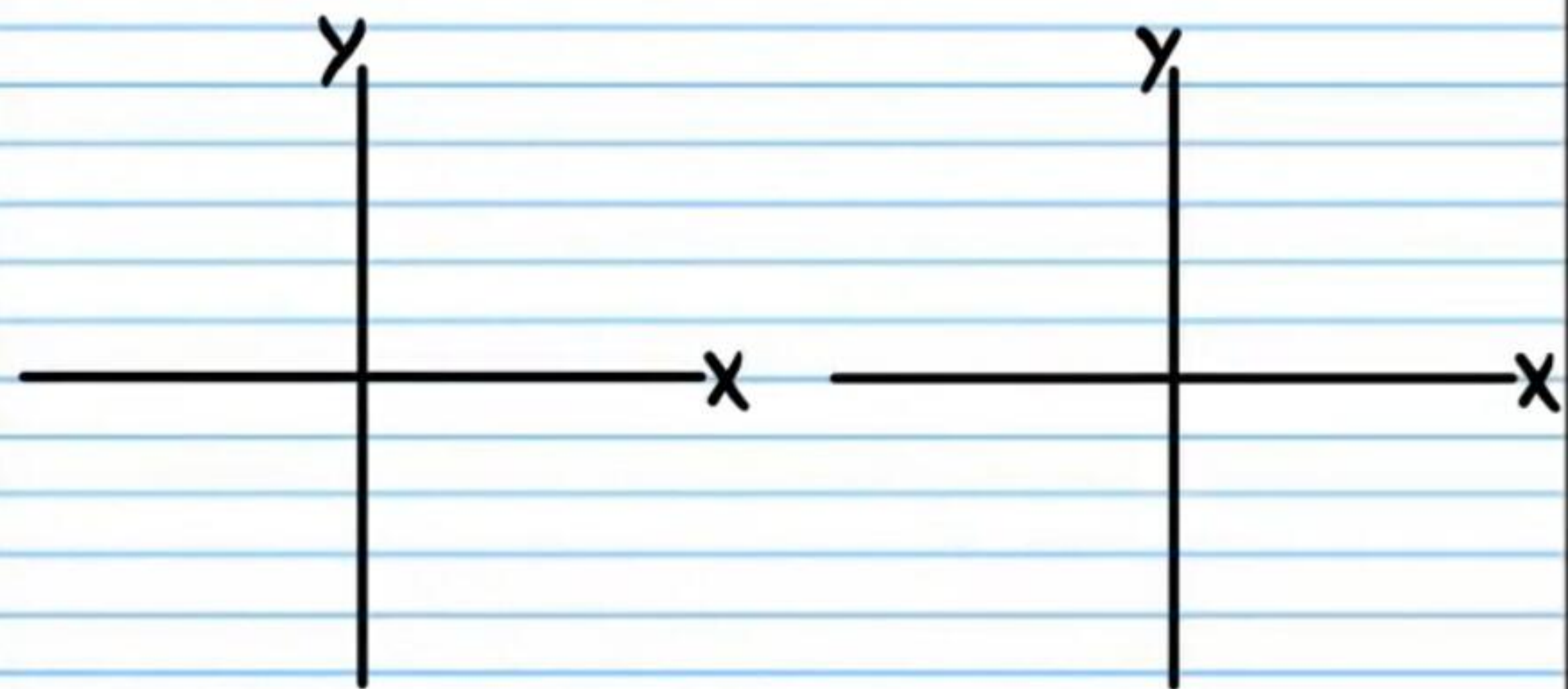


⑫ If  $\cos \theta = \frac{1}{\sqrt{2}}$ , find all values of  $\theta$ .



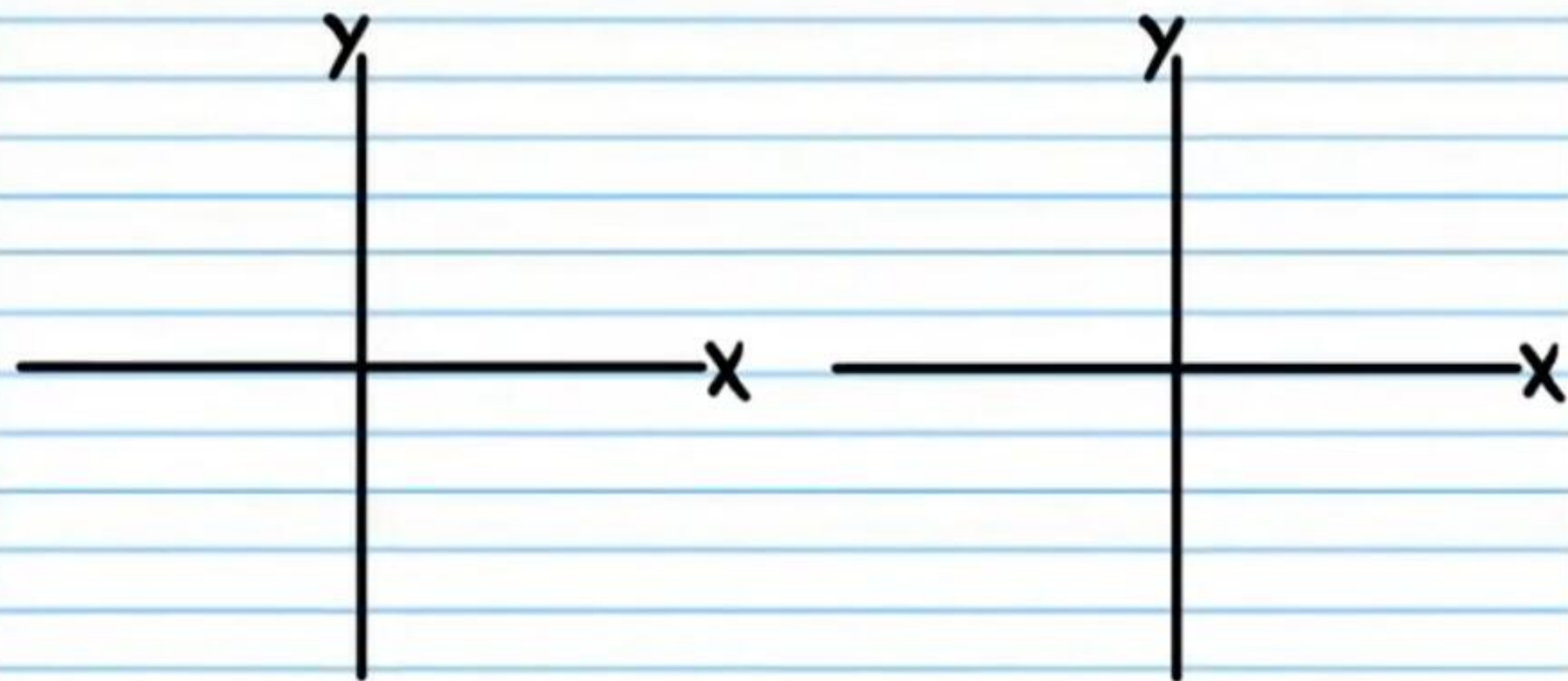
For problem #'s 19-21, use a calculator to find the degree measures. Draw each angle in standard position, the corresponding right triangle, and the reference angle.

⑲ If  $\cos \theta = \frac{11}{12}$ , and  $0^\circ < \theta \leq 360^\circ$ , find the values of  $\theta$ .

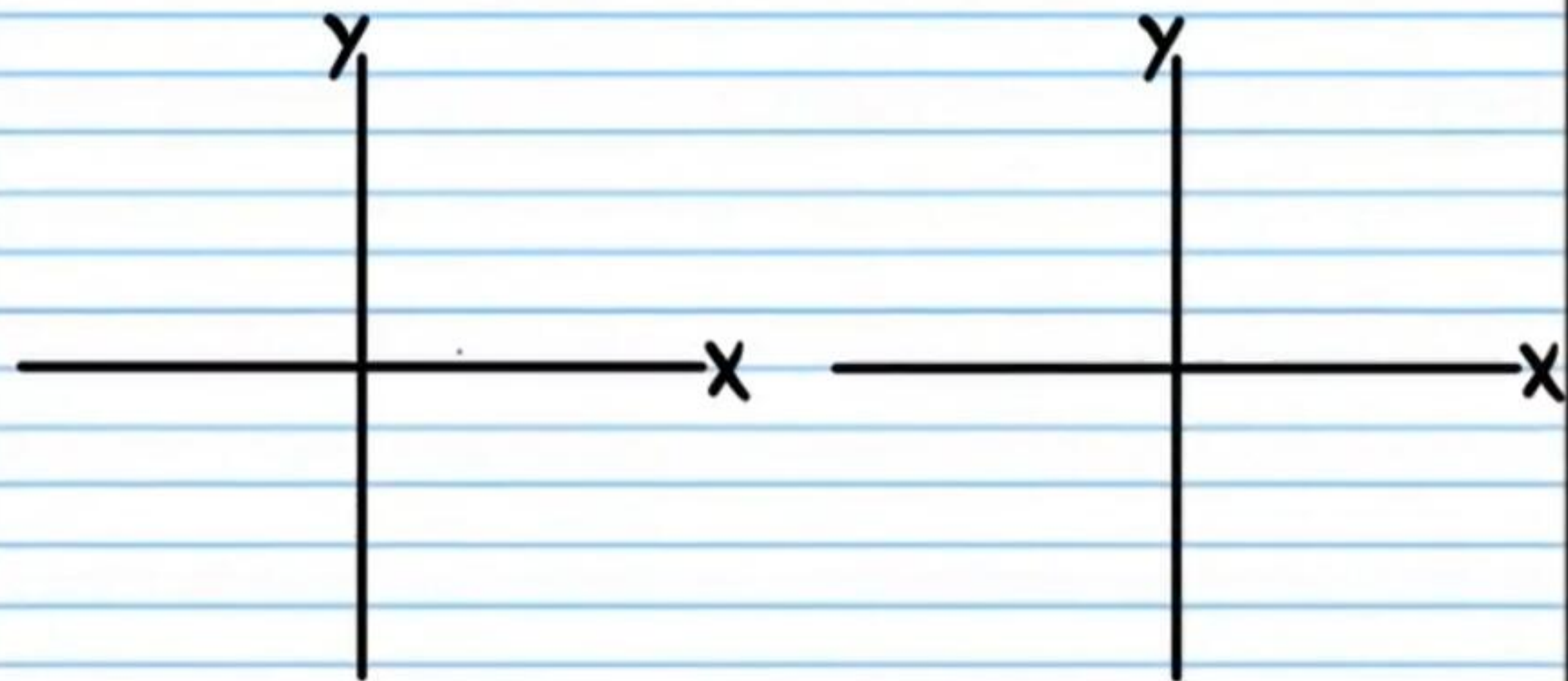




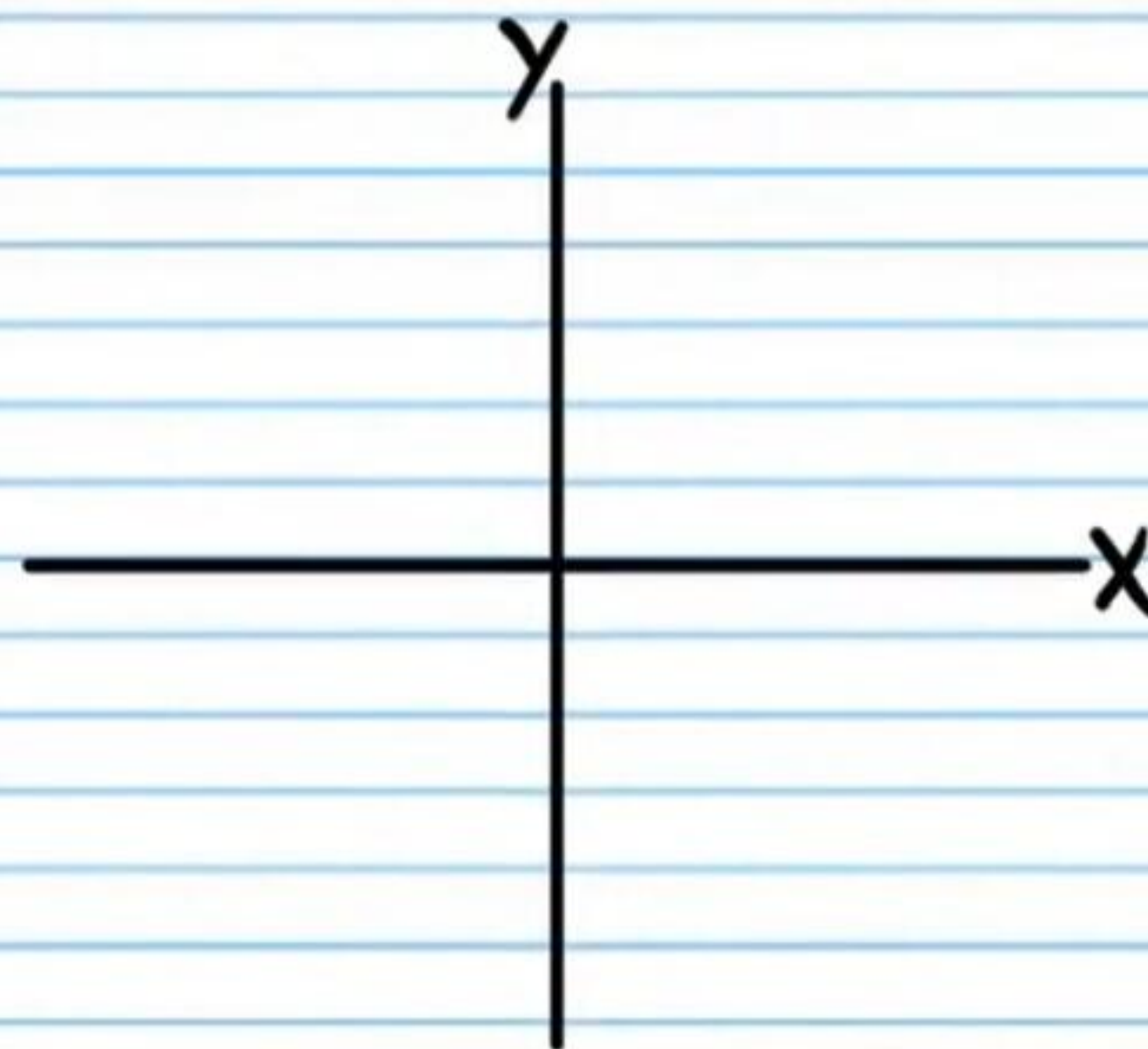
②① If  $\sin \theta = \frac{1}{3}$ , and  $0^\circ < \theta \leq 360^\circ$ , Find the values of  $\theta$ .



②① If  $\tan \theta = -3$ , and  $0^\circ < \theta \leq 360^\circ$ , Find the values of  $\theta$ .



②② If  $\cos \beta = -\frac{1}{\sqrt{3}}$ , and  $\beta$  terminates in quadrant III, find the values of the five remaining basic trigonometric functions of  $\beta$ .



i)  $\cot \beta$

ii)  $\sec \beta$

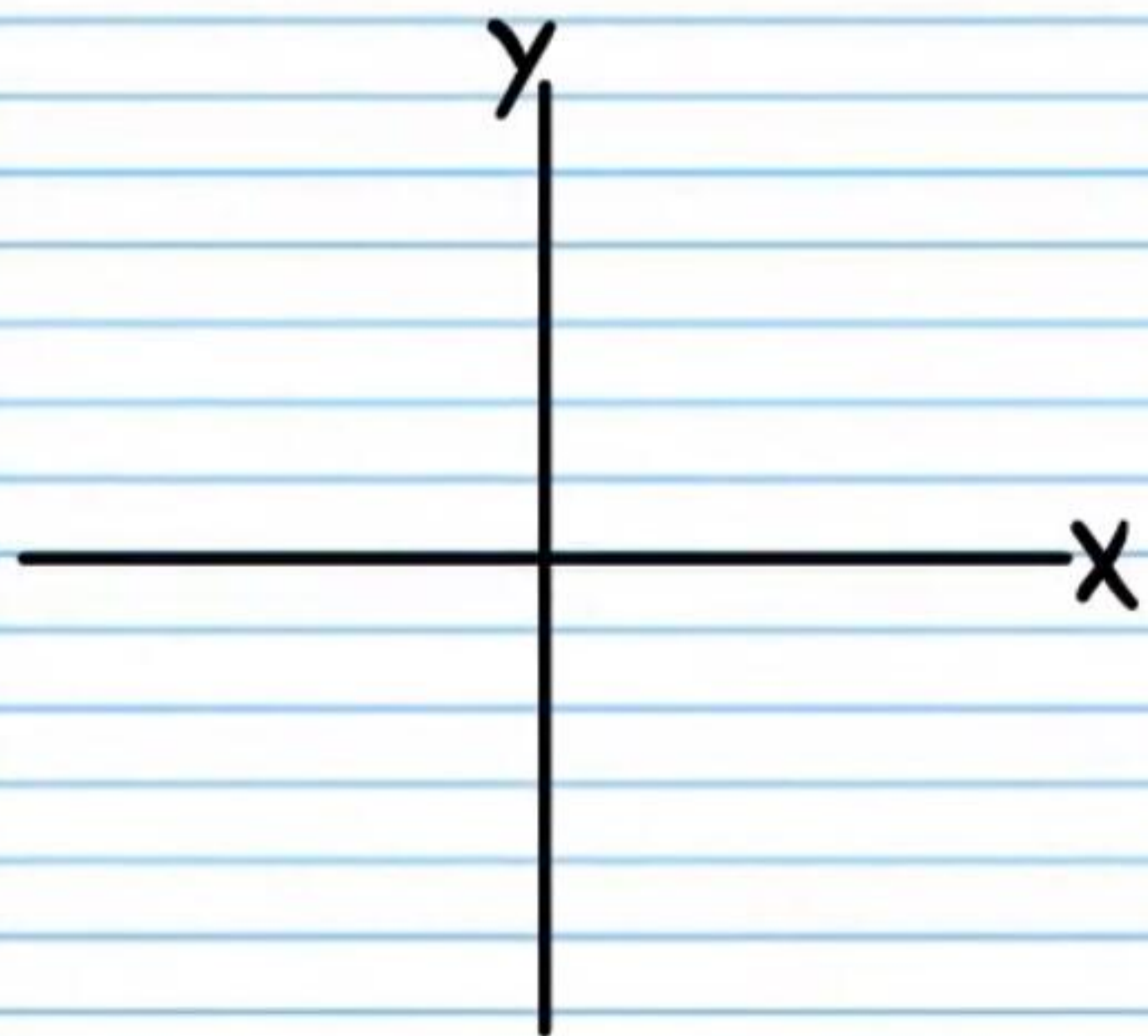
iii)  $\sin \beta$

iv)  $\csc \beta$

v)  $\tan \beta$



23) If  $\tan W = 2$ , and  $W$  terminates in quadrant I, find the values of the five remaining basic trigonometric functions of  $W$ .



i)  $\csc W$

ii)  $\sec W$

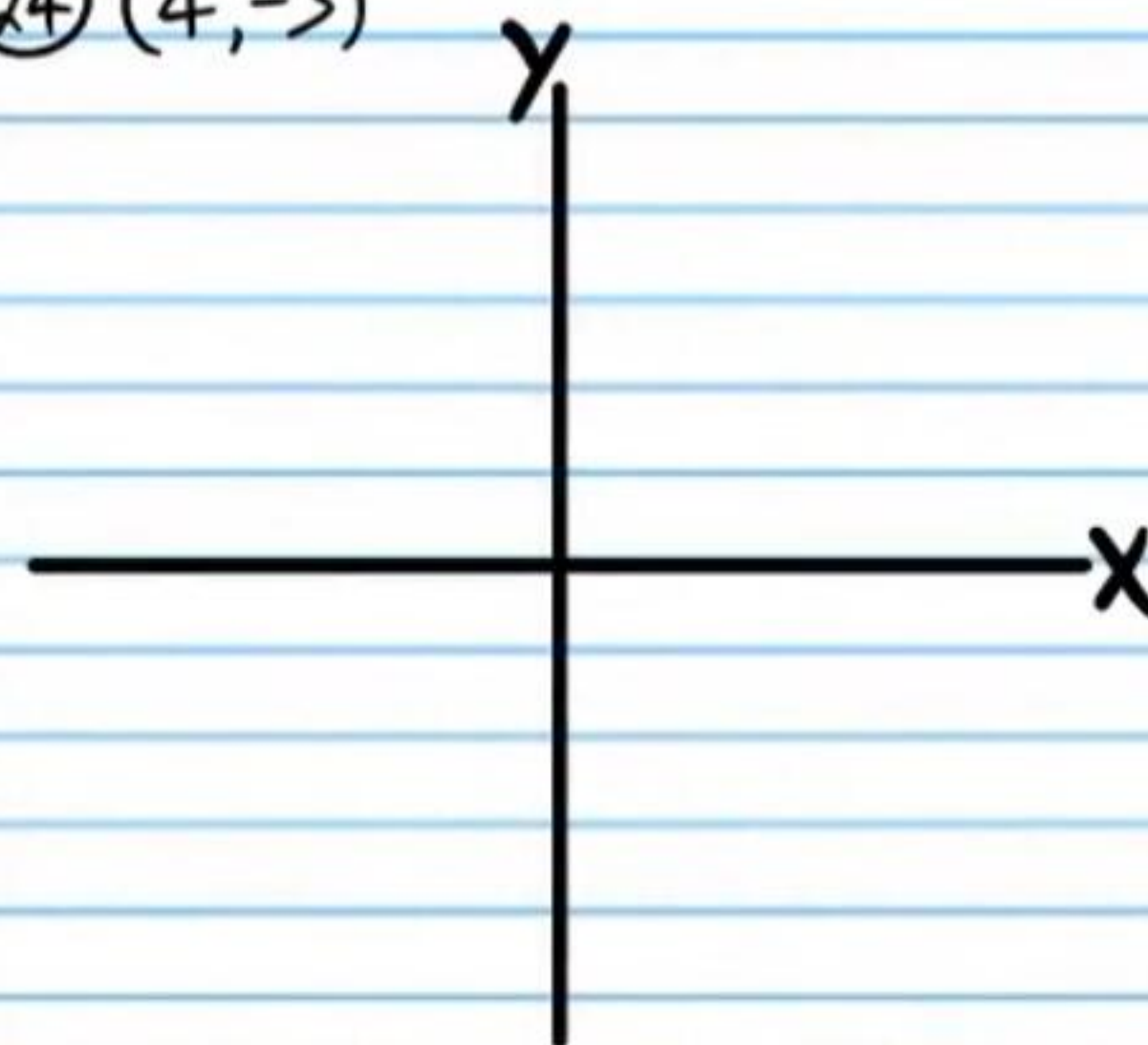
iii)  $\cot W$

iv)  $\cos W$

v)  $\sin W$

The coordinates of points are given below. If each of these points lies on the terminal side of an angle in standard position, find the values of the six basic trigonometric functions of the angle.

24)  $(4, -3)$



i)  $\sin \theta$

ii)  $\cos \theta$

iii)  $\tan \theta$

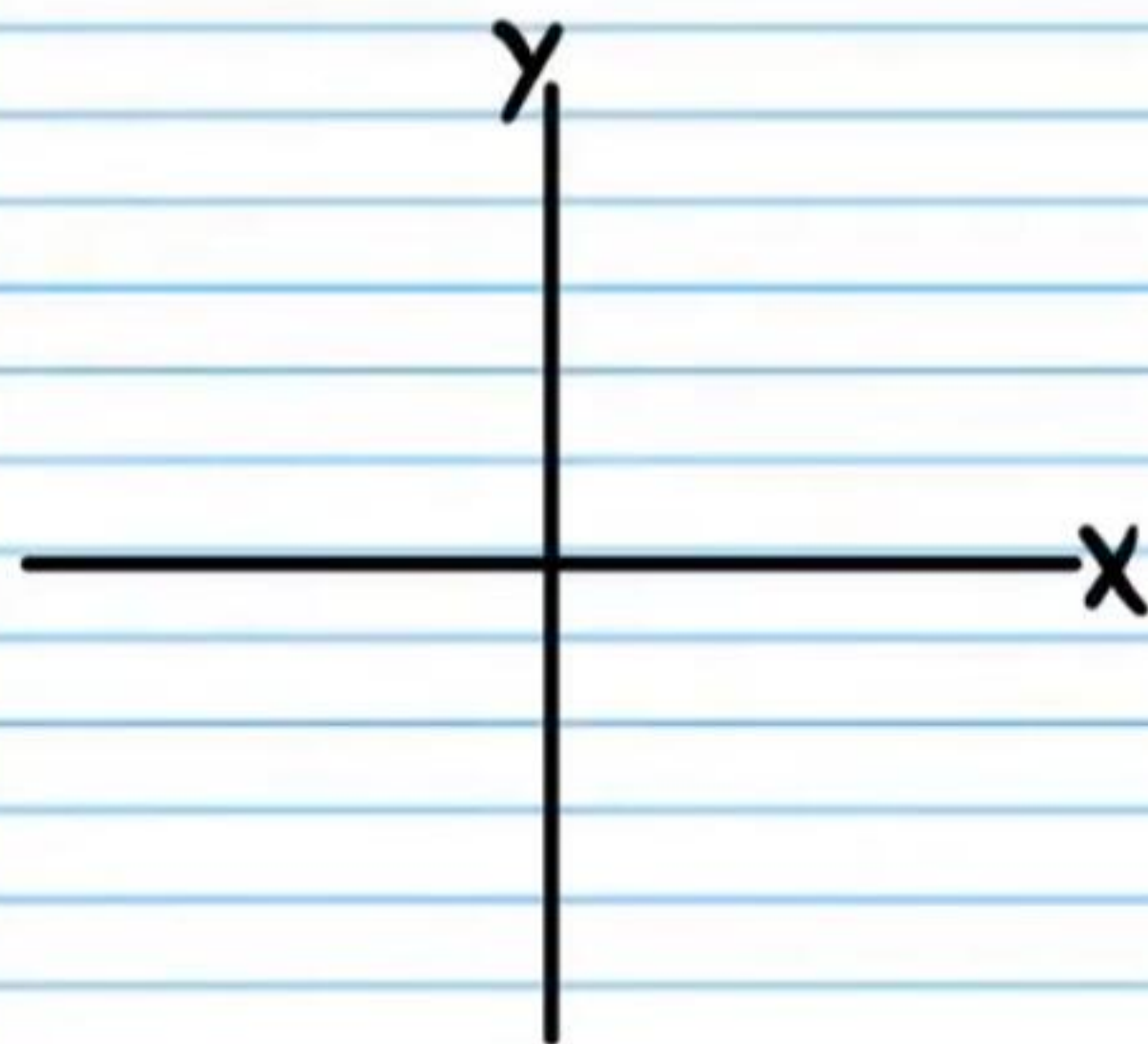
iv)  $\csc \theta$

v)  $\sec \theta$

vi)  $\cot \theta$



Q5)  $(-2, 2)$



i)  $\sin \theta$

ii)  $\cos \theta$

iii)  $\tan \theta$

iv)  $\csc \theta$

v)  $\sec \theta$

vi)  $\cot \theta$



# Practice Test I (Version 2) Answers

Name  
Date  
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① Possible answers:  $-589^\circ$ ,  $-229^\circ$ ,  $491^\circ$ ,  $851^\circ$ ,  $1211^\circ$ . Only three are needed.

②  $\theta = 0^\circ, 360^\circ$

③  $\theta = 180n$ ,  $n$  is any integer.

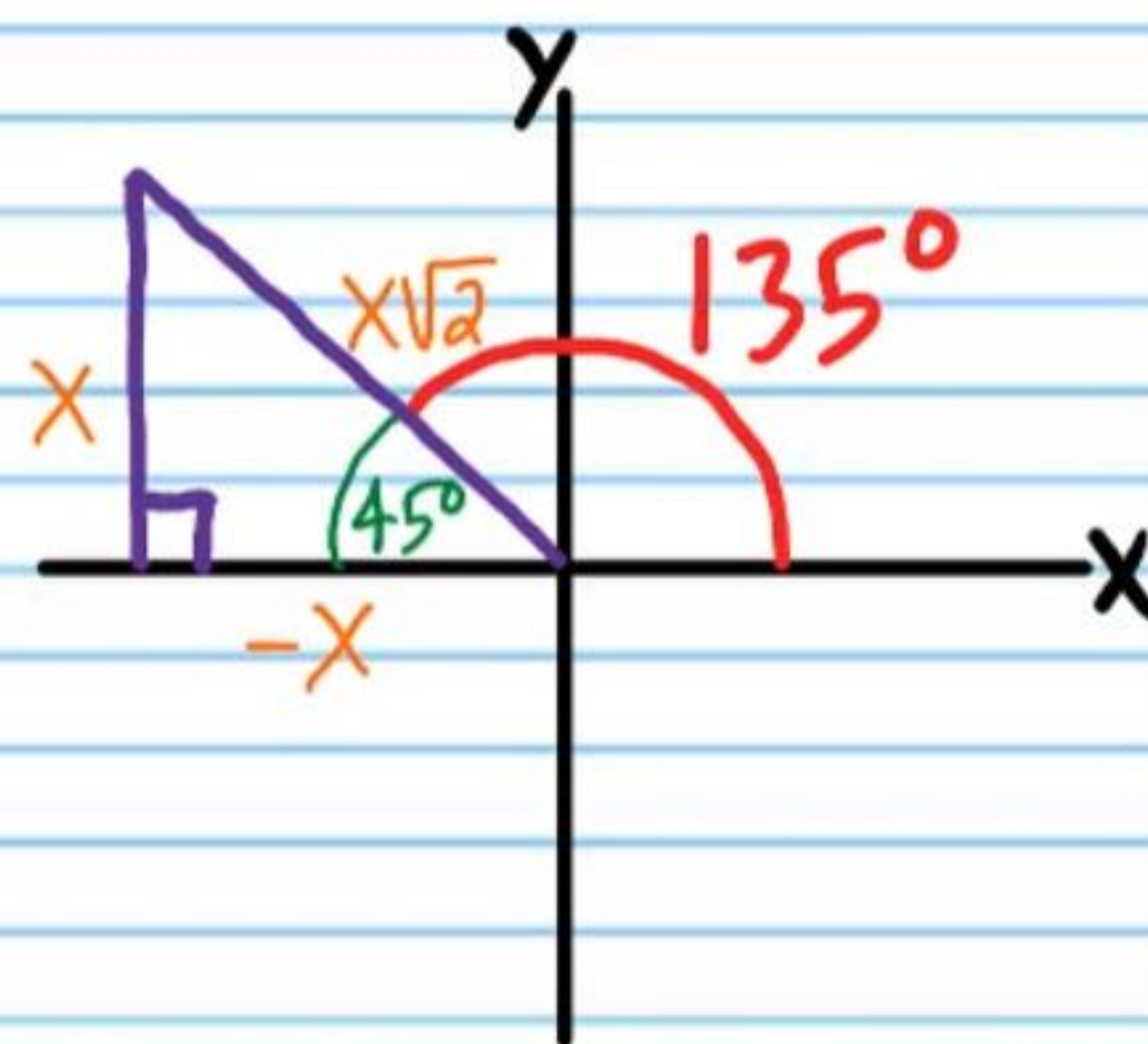
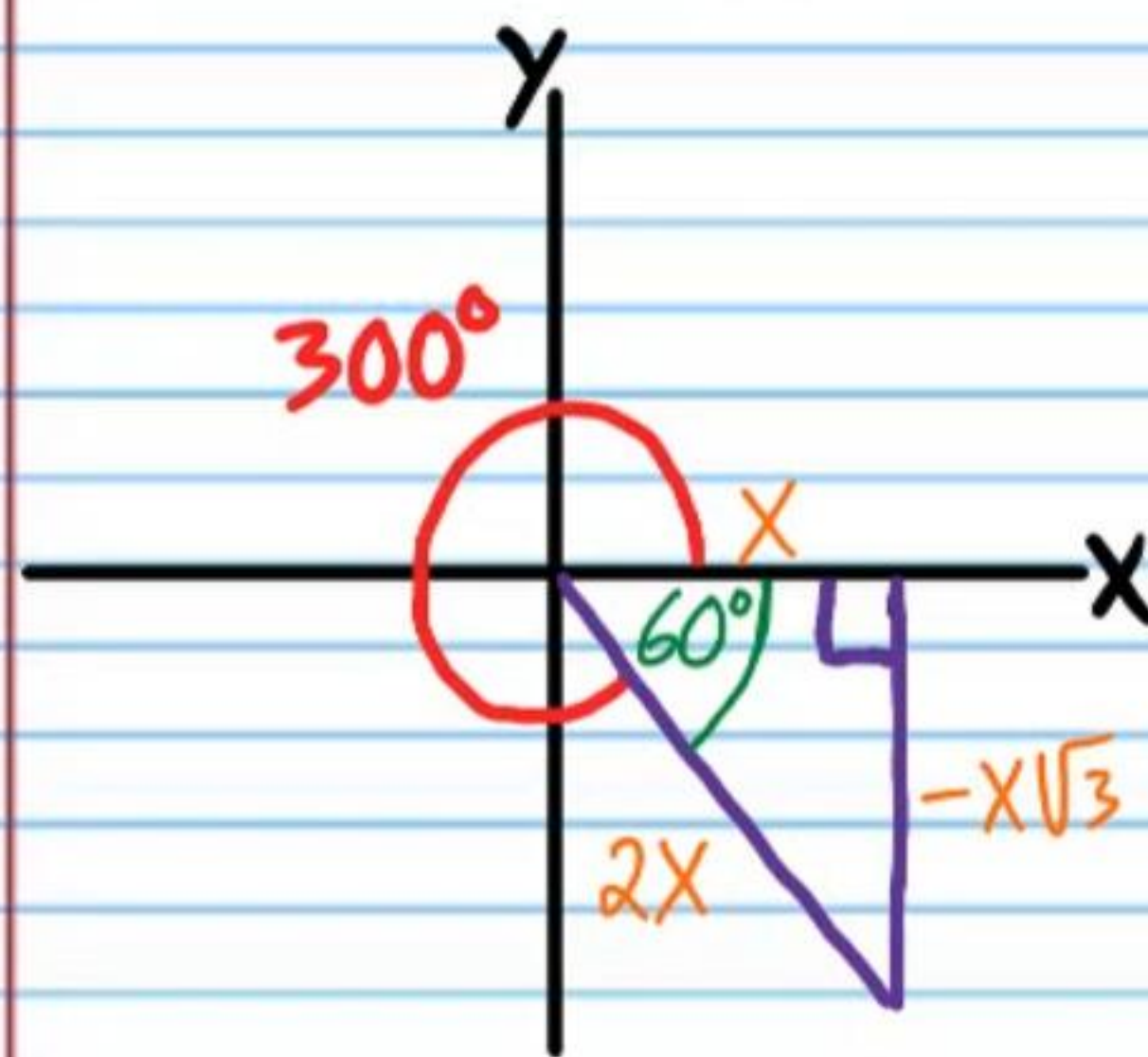
④  $\theta = 180^\circ$

⑤ i)  $-0.951$

ii)  $6.314$

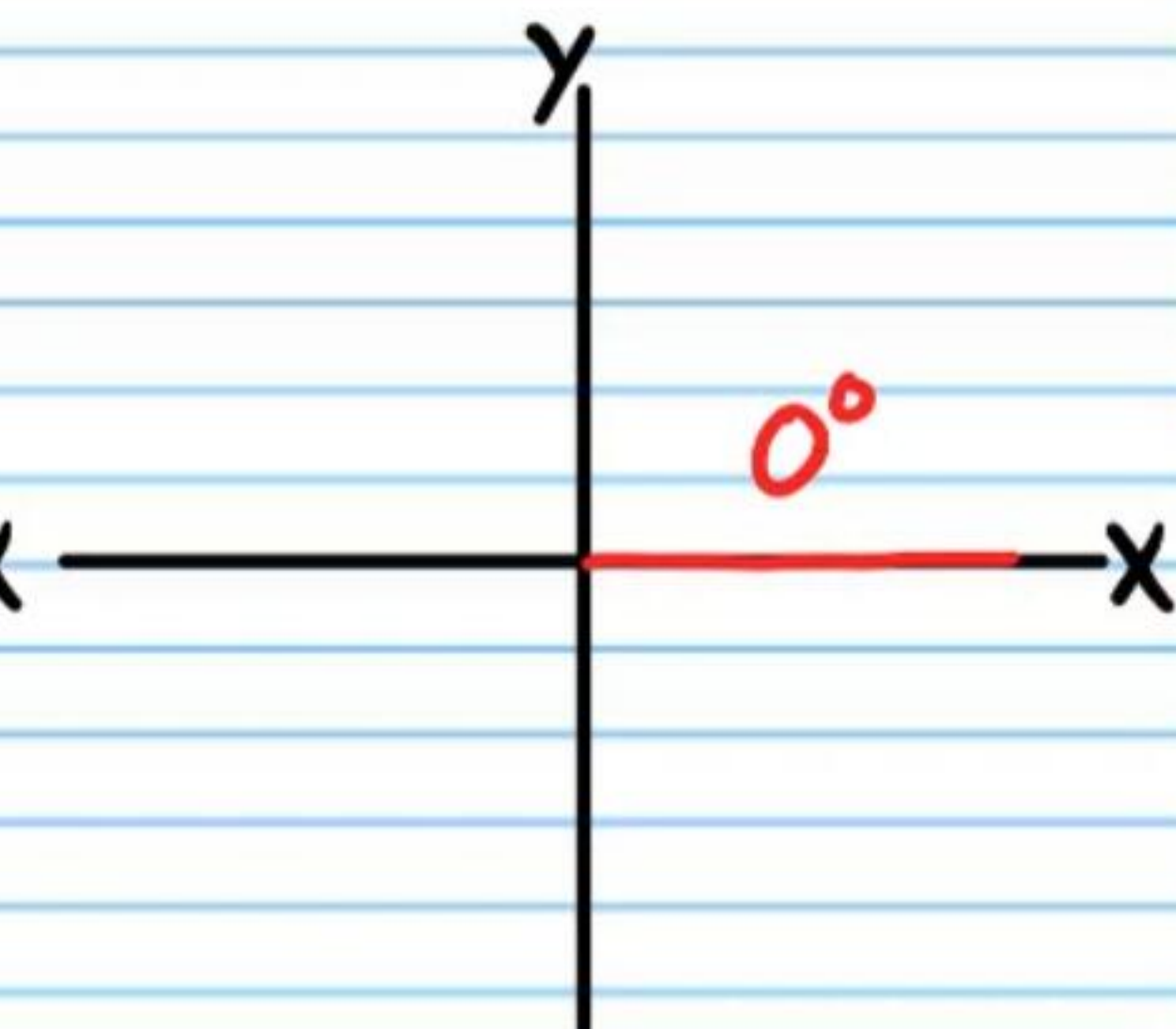
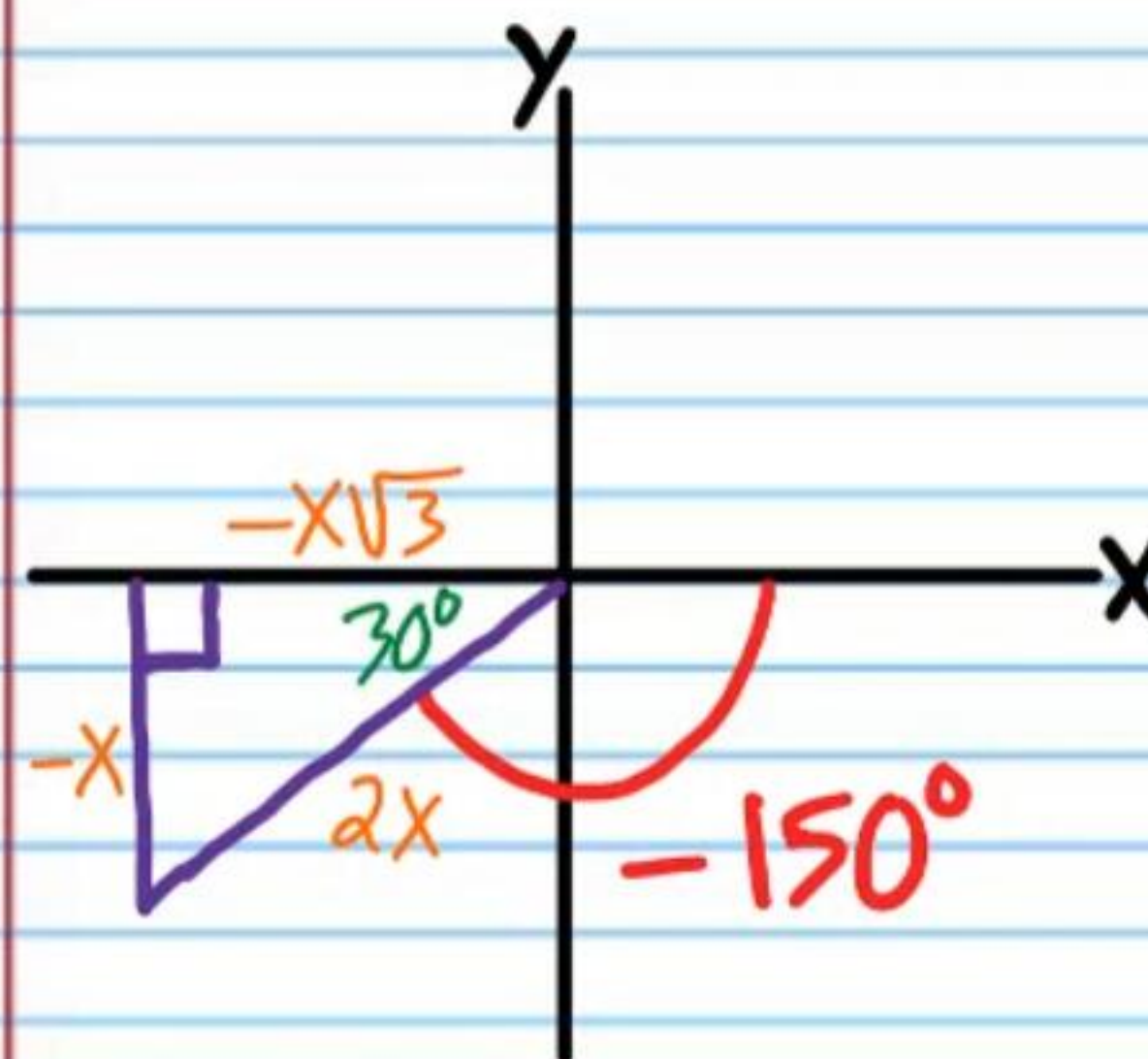
⑥  $\tan 300^\circ = -\sqrt{3}$

⑦  $\cot 135^\circ = -1$



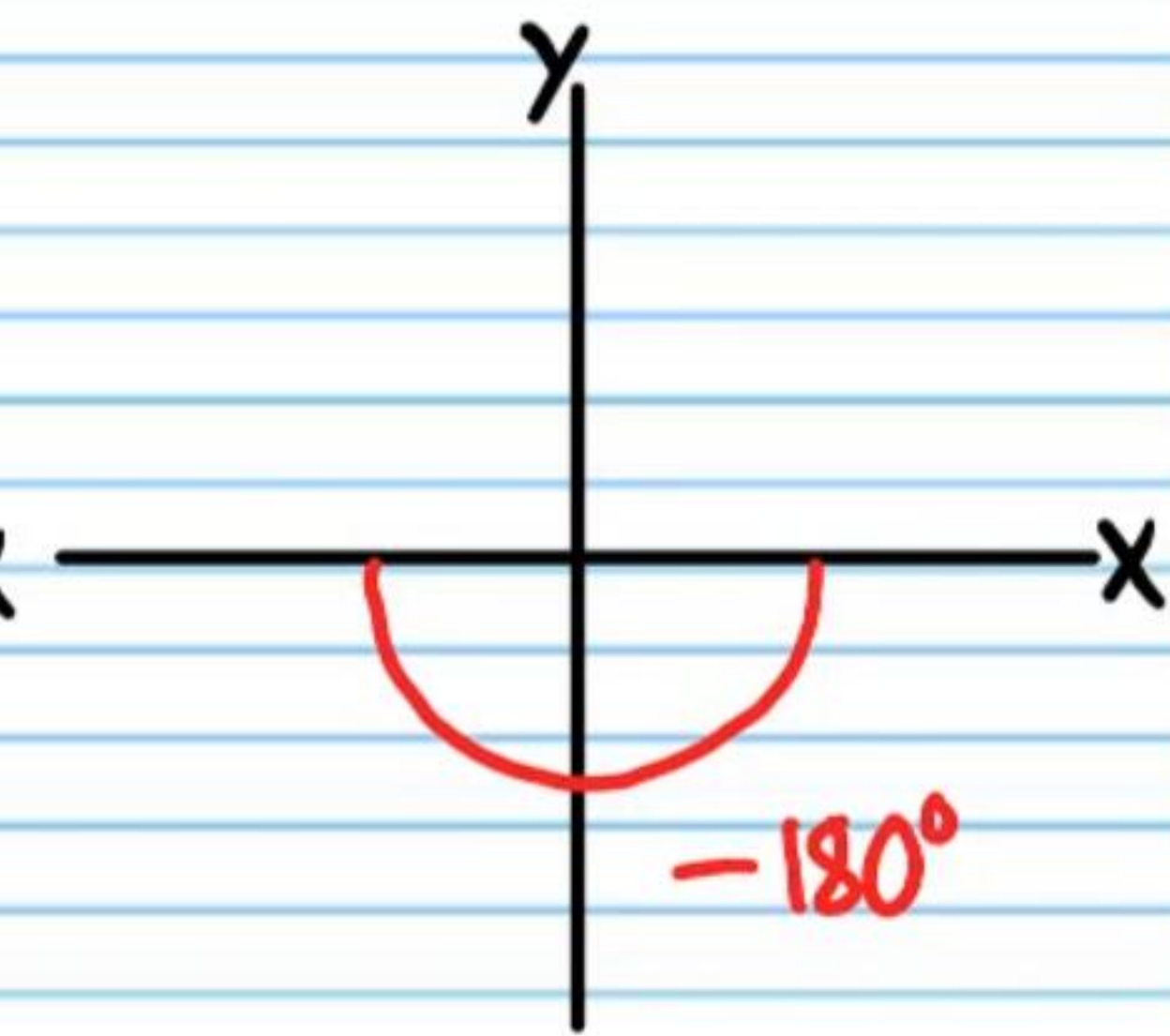
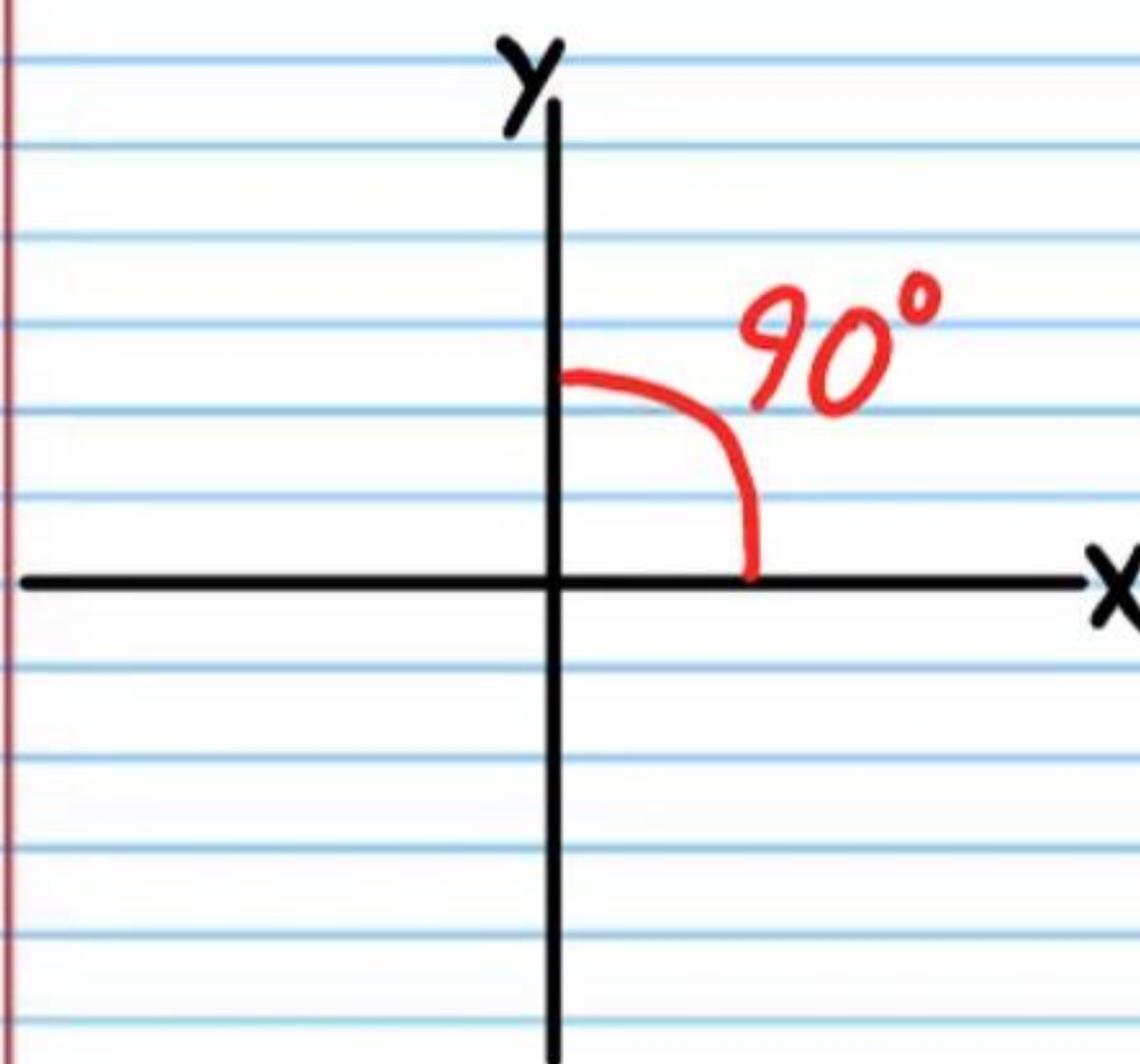
⑧  $\cos(-150^\circ) = -\frac{\sqrt{3}}{2}$

⑨  $\csc 0^\circ = \text{Undefined}$



⑩  $\sin 90^\circ = 1$

⑪  $\sec(-180^\circ) = -1$

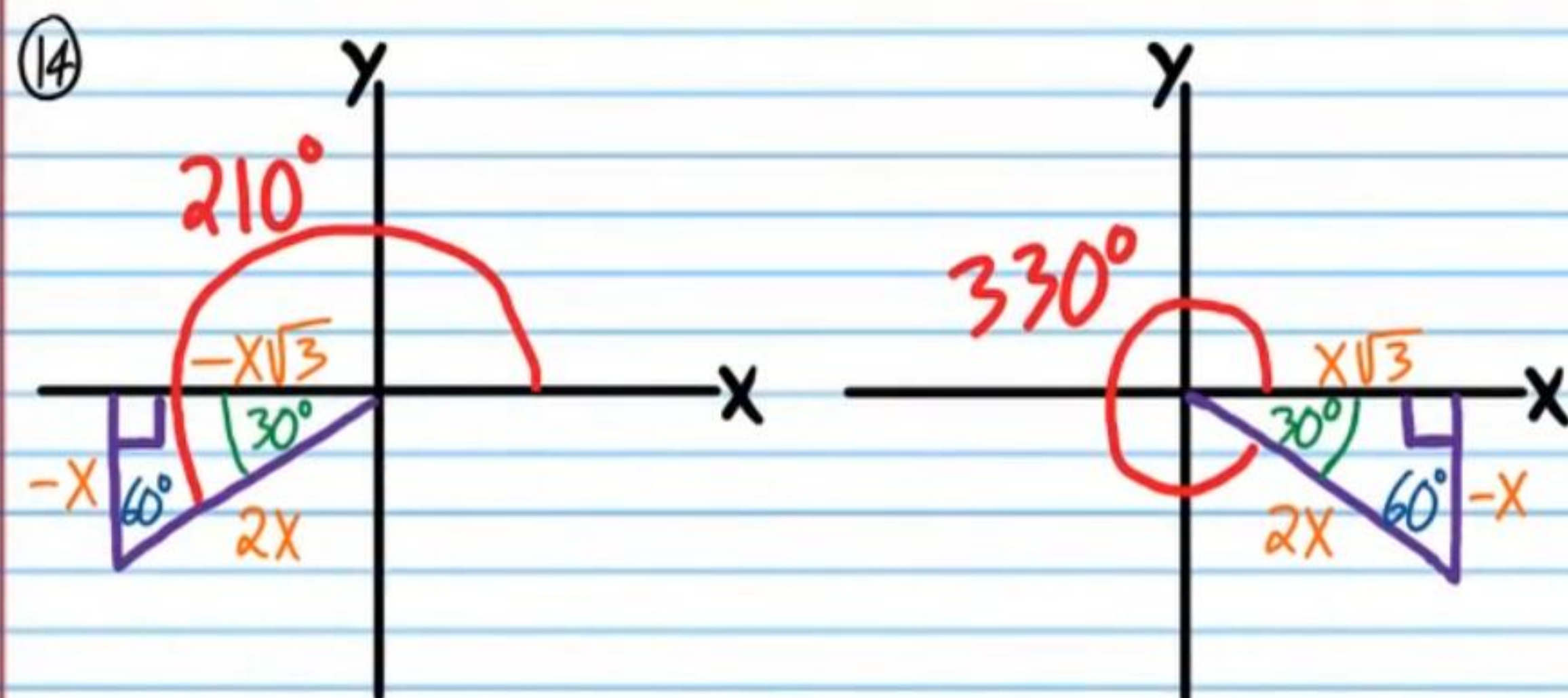
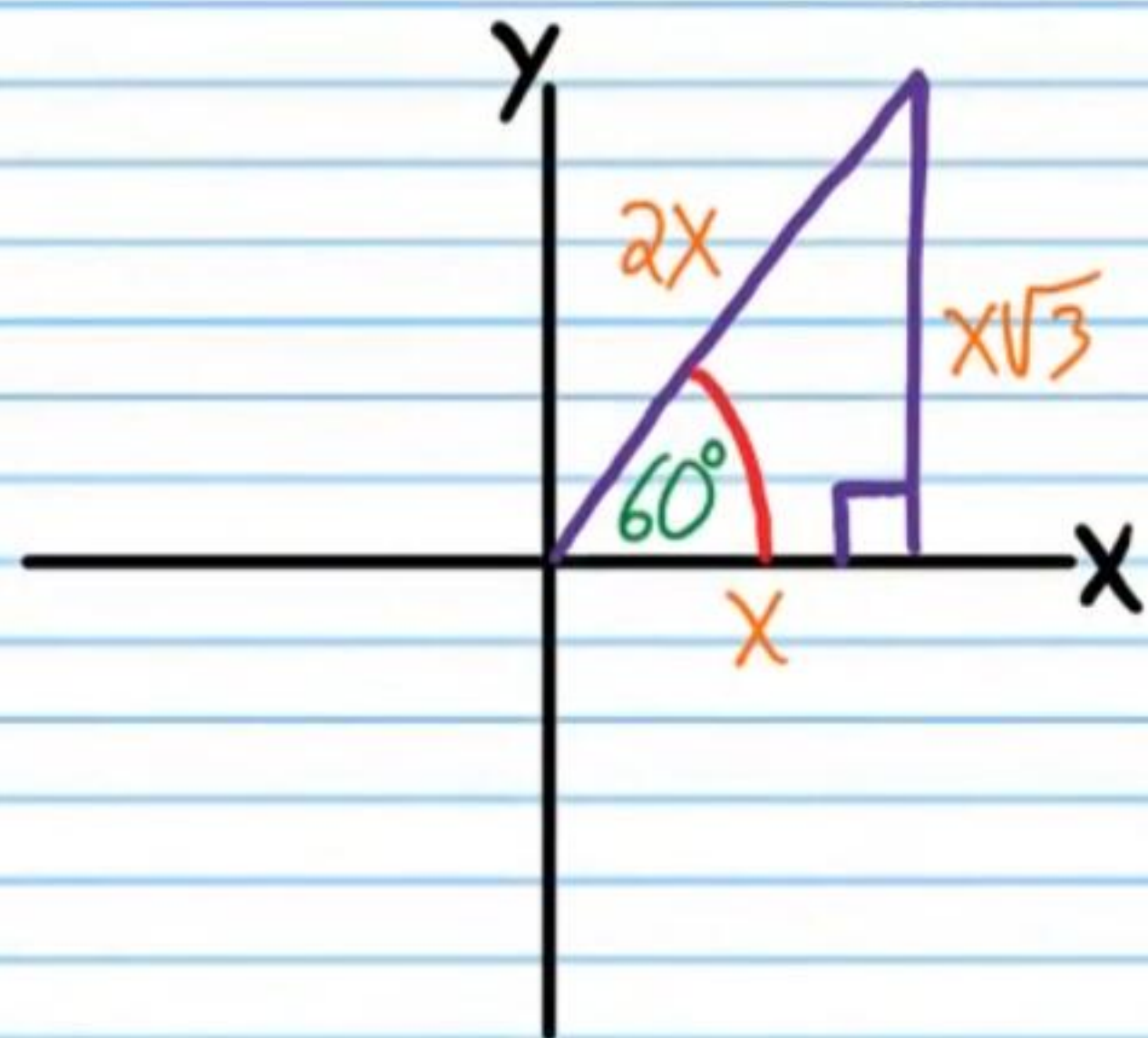




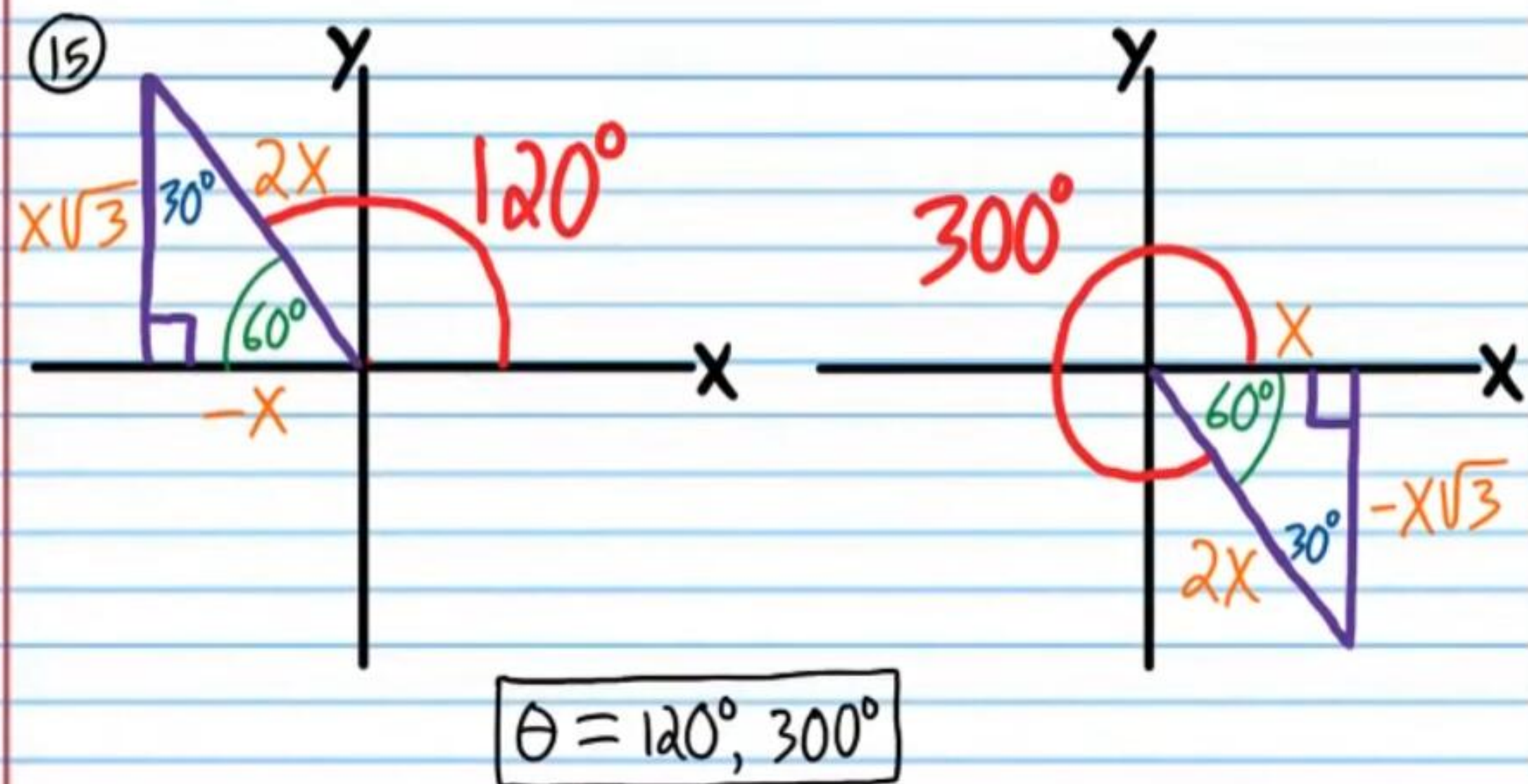
$$(12) \csc 60^\circ = \frac{2}{\sqrt{3}} = \boxed{\frac{2\sqrt{3}}{3}}$$

$$(13) \text{ i) } \frac{2}{\sqrt{13}} = \boxed{\frac{2\sqrt{13}}{13}}$$

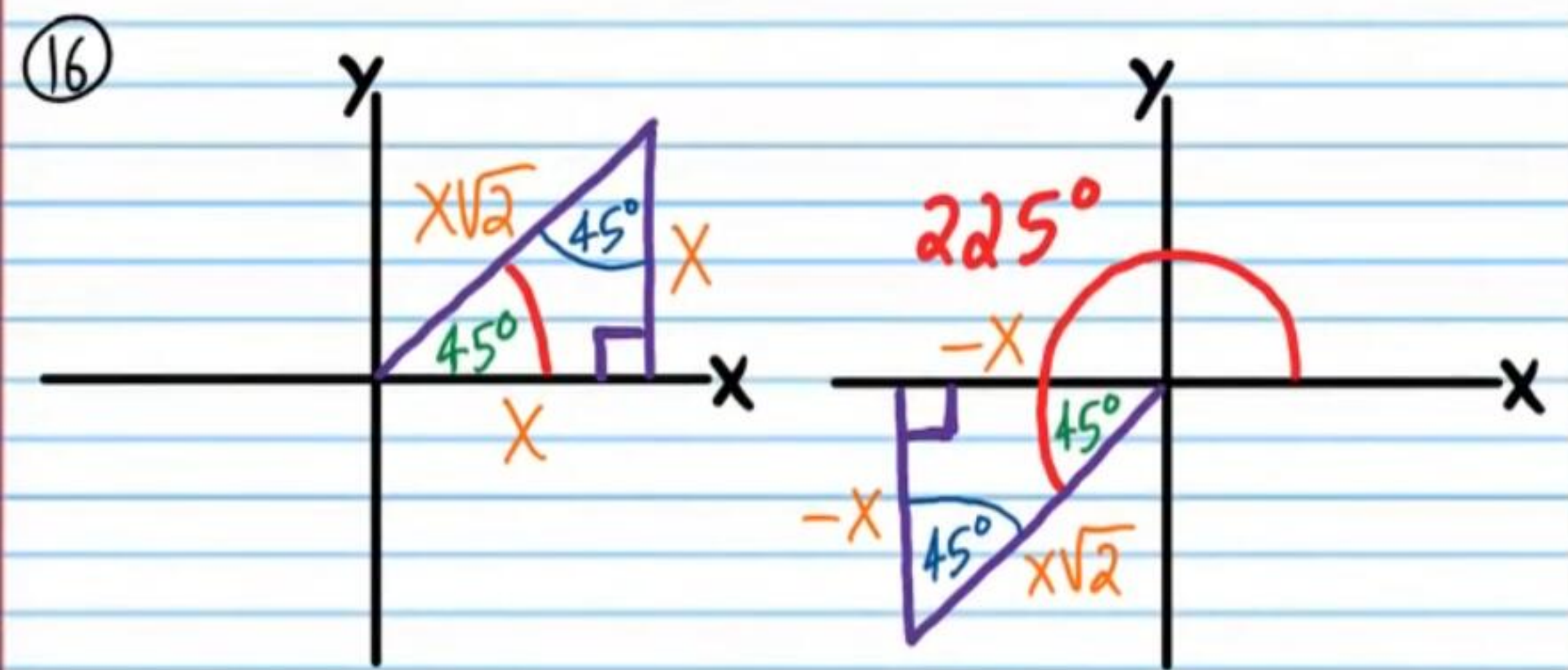
$$\text{ii) } \boxed{\frac{17}{8}}$$



$$\theta = 210^\circ, 330^\circ$$



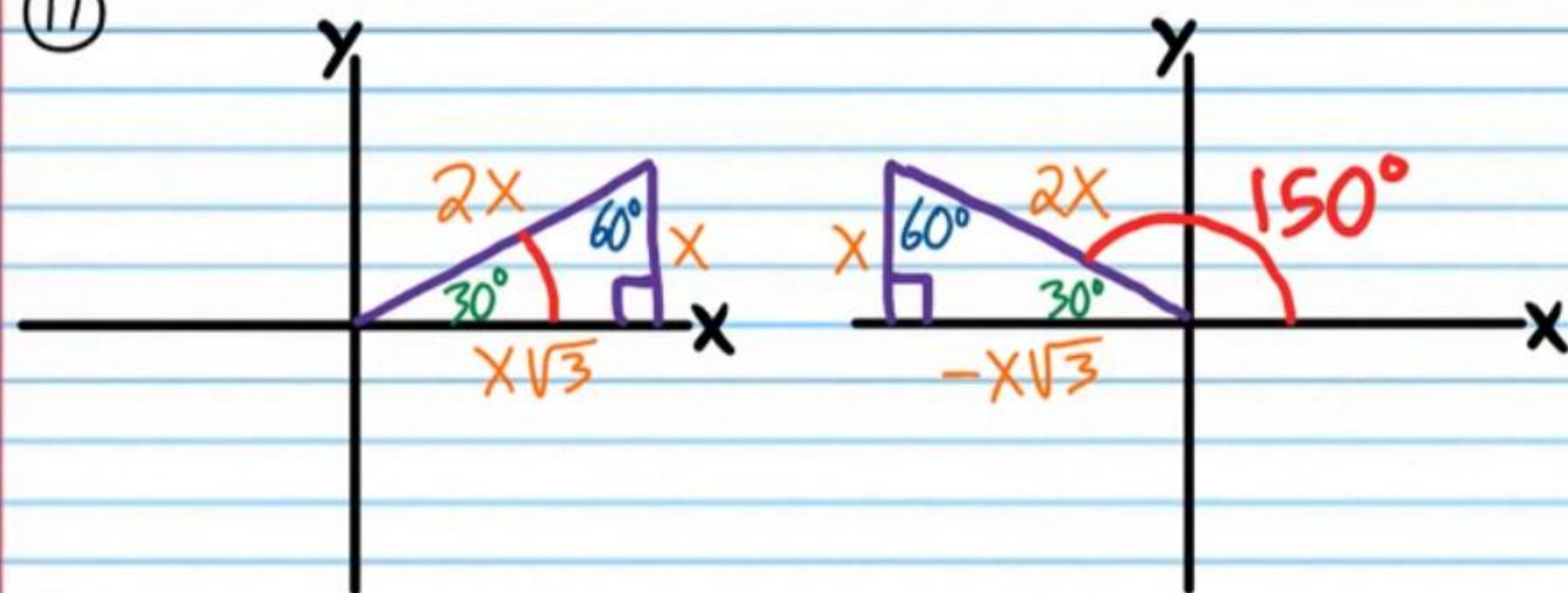
$$\theta = 120^\circ, 300^\circ$$



$$\theta = 45^\circ, 225^\circ$$



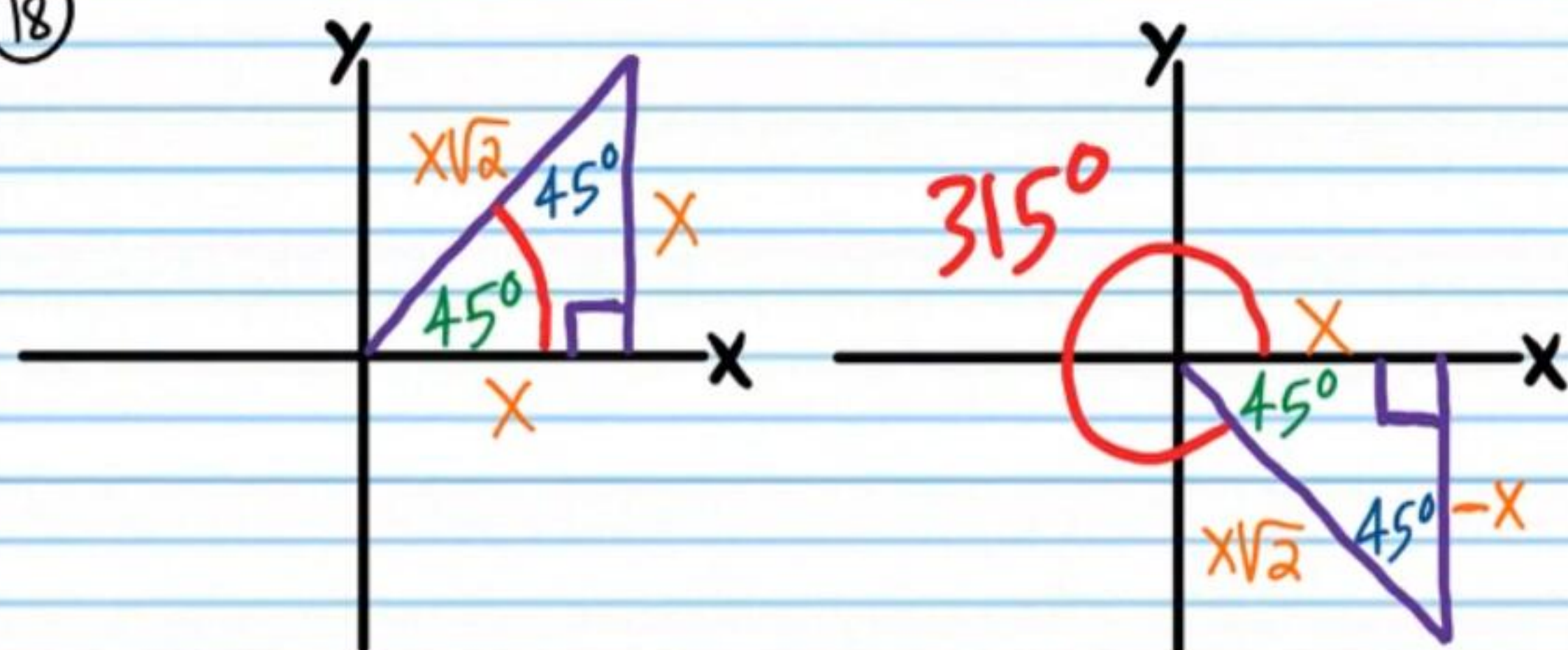
17



$$\theta = 30^\circ + 360n, n \text{ is any integer.}$$

$$\theta = 150^\circ + 360n, n \text{ is any integer.}$$

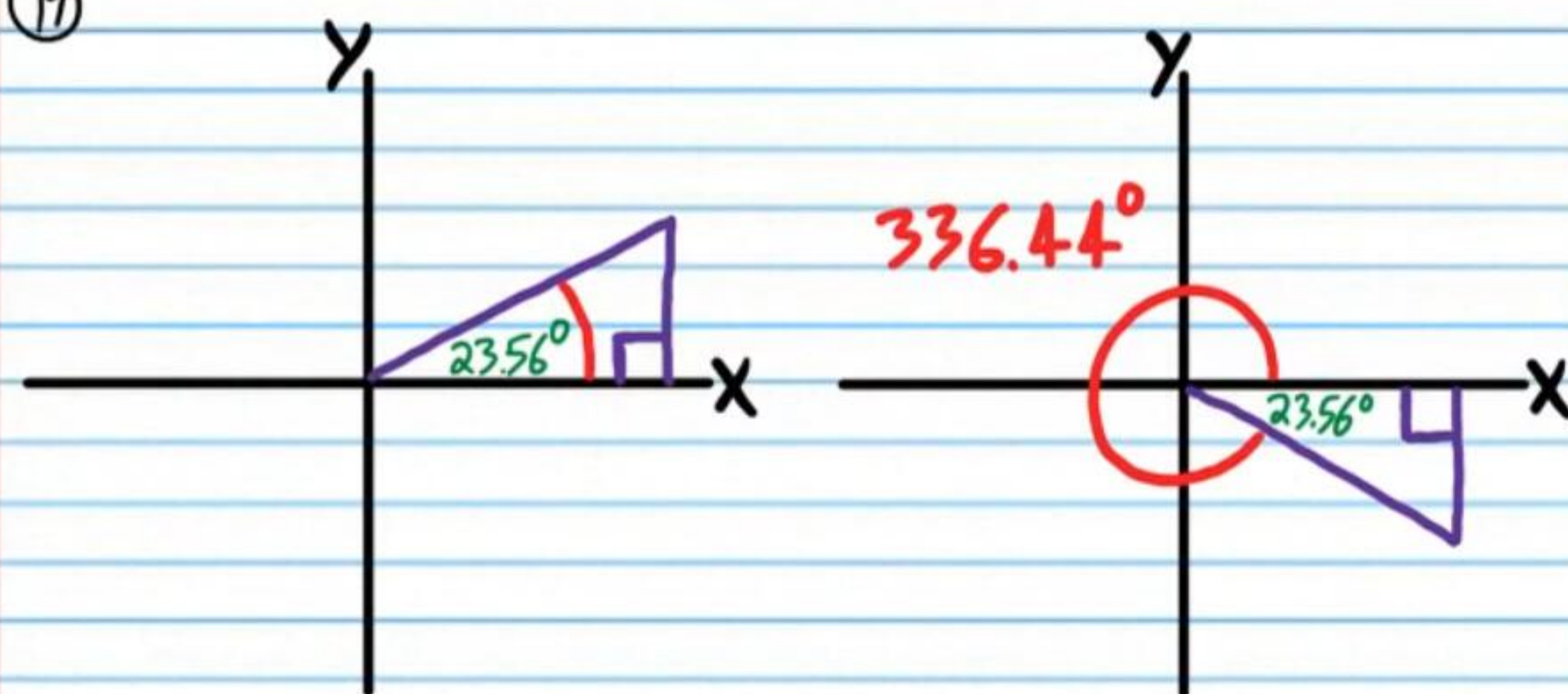
18



$$\theta = 45^\circ + 360n, n \text{ is any integer.}$$

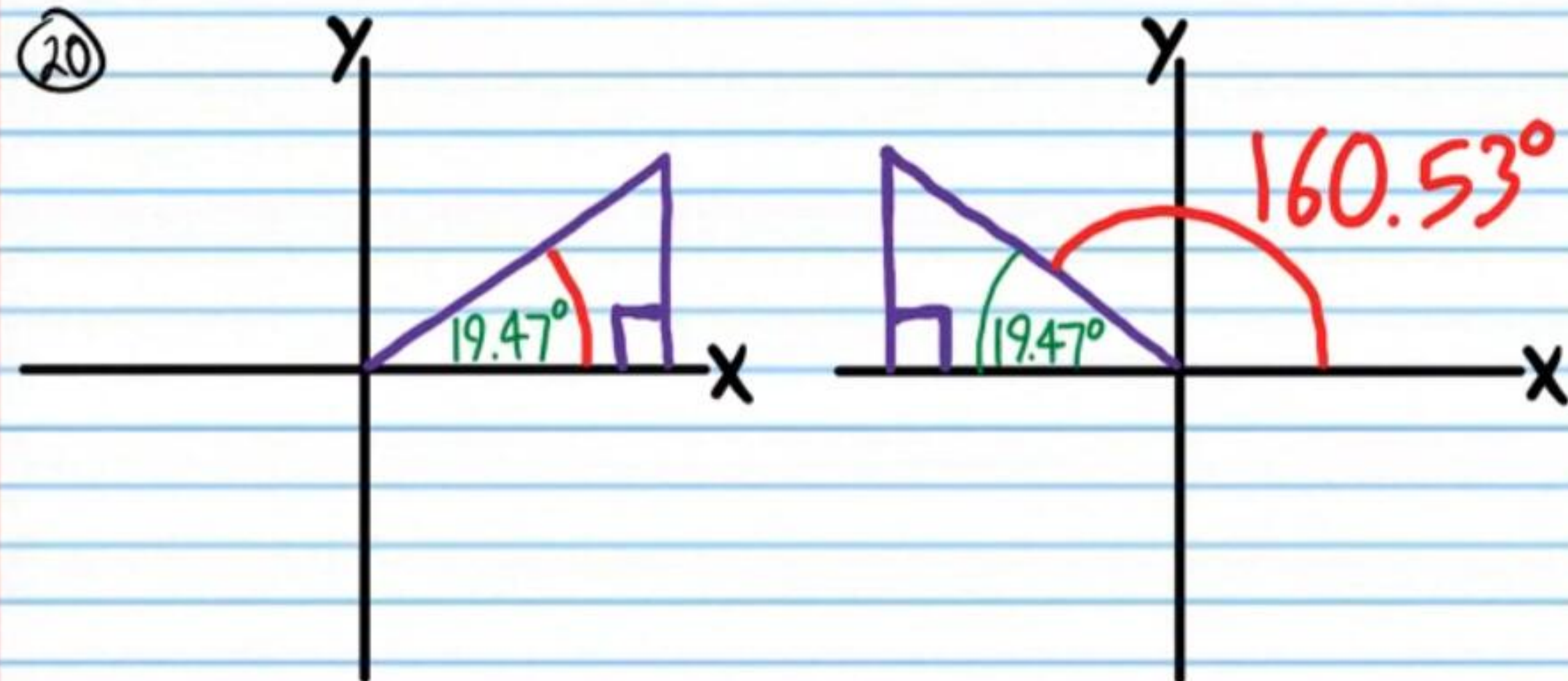
$$\theta = 315^\circ + 360n, n \text{ is any integer.}$$

19



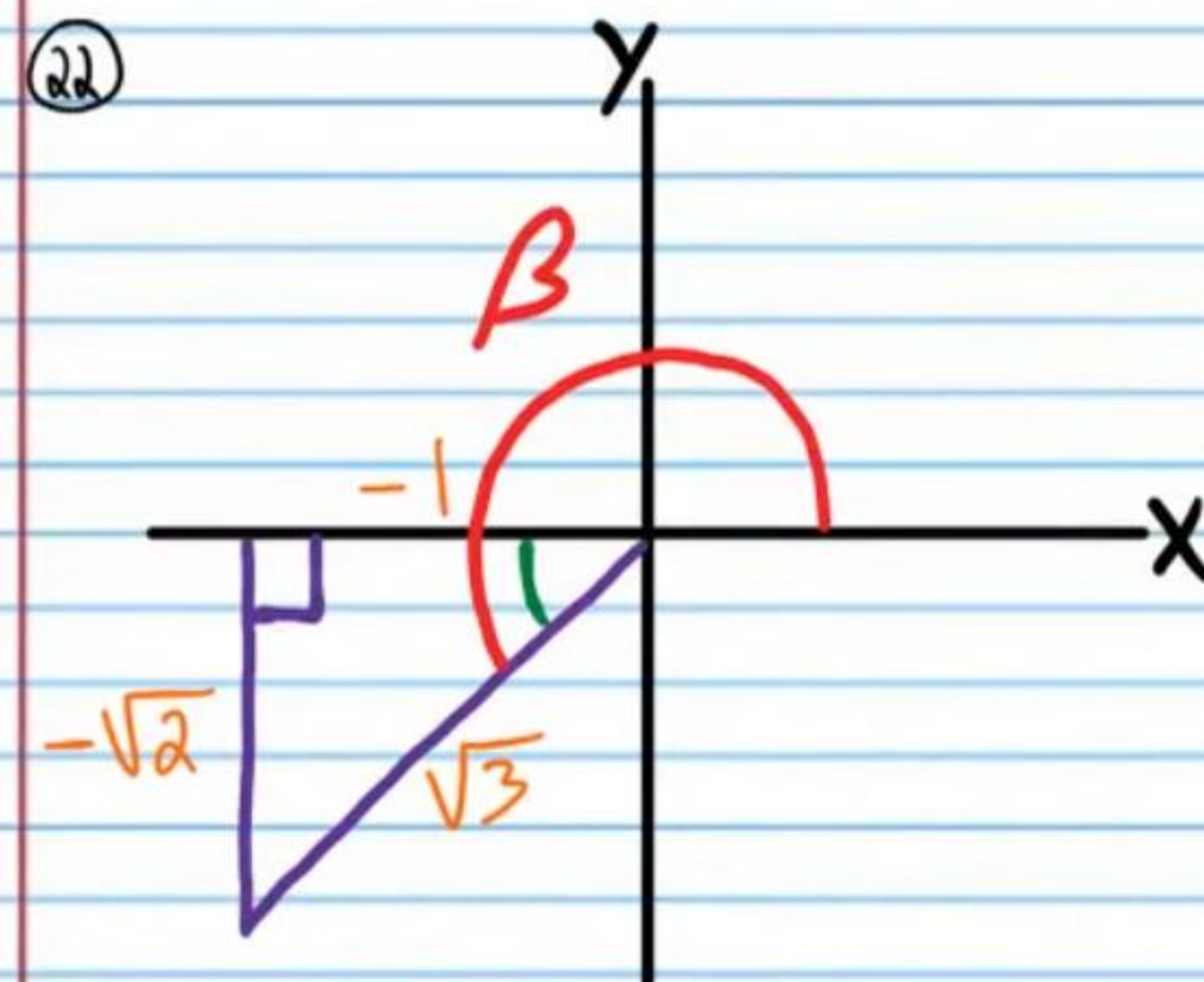
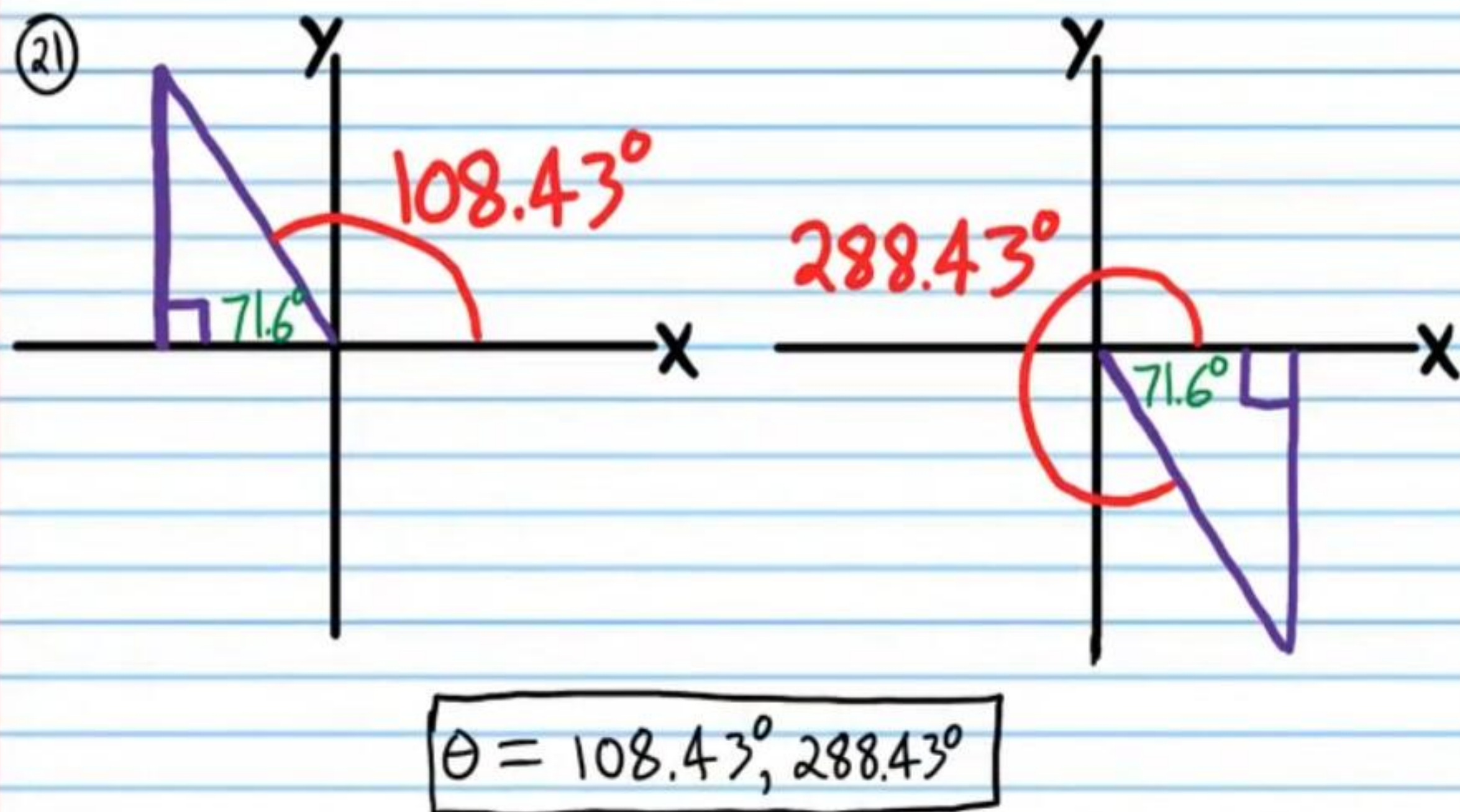
$$\theta = 23.56^\circ, 336.44^\circ$$

20



$$\theta = 19.47^\circ, 160.53^\circ$$





i)  $\cot \beta = \frac{1}{\sqrt{2}} = \boxed{\frac{\sqrt{2}}{2}}$

ii)  $\sec \beta = \boxed{-\sqrt{3}}$

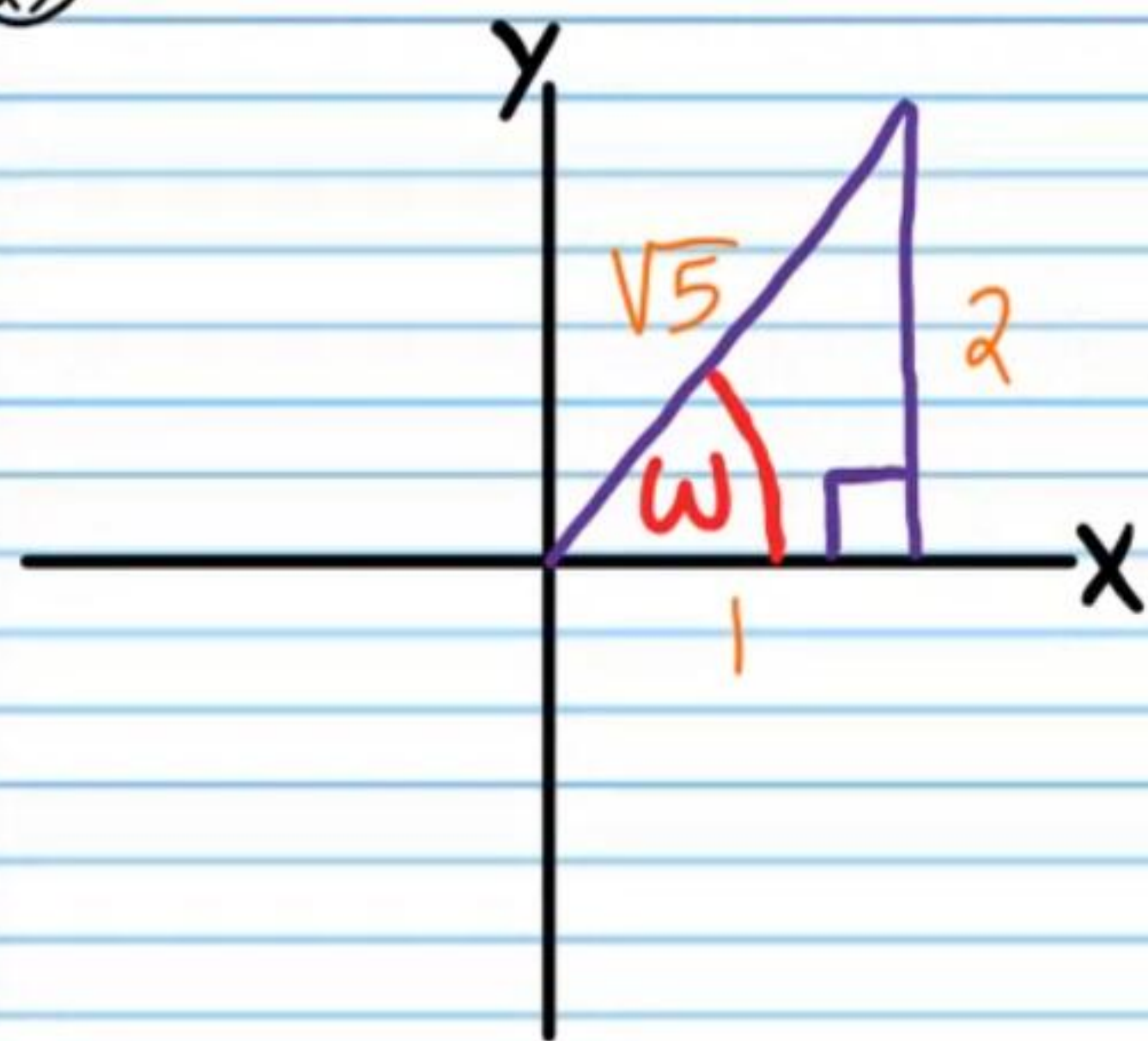
iii)  $\sin \beta = -\frac{\sqrt{2}}{\sqrt{3}} = \boxed{-\frac{\sqrt{6}}{3}}$

iv)  $\csc \beta = -\frac{\sqrt{3}}{\sqrt{2}} = \boxed{-\frac{\sqrt{6}}{2}}$

v)  $\tan \beta = \boxed{\sqrt{2}}$



(23)



$$\text{i) } \csc w = \boxed{\frac{\sqrt{5}}{2}}$$

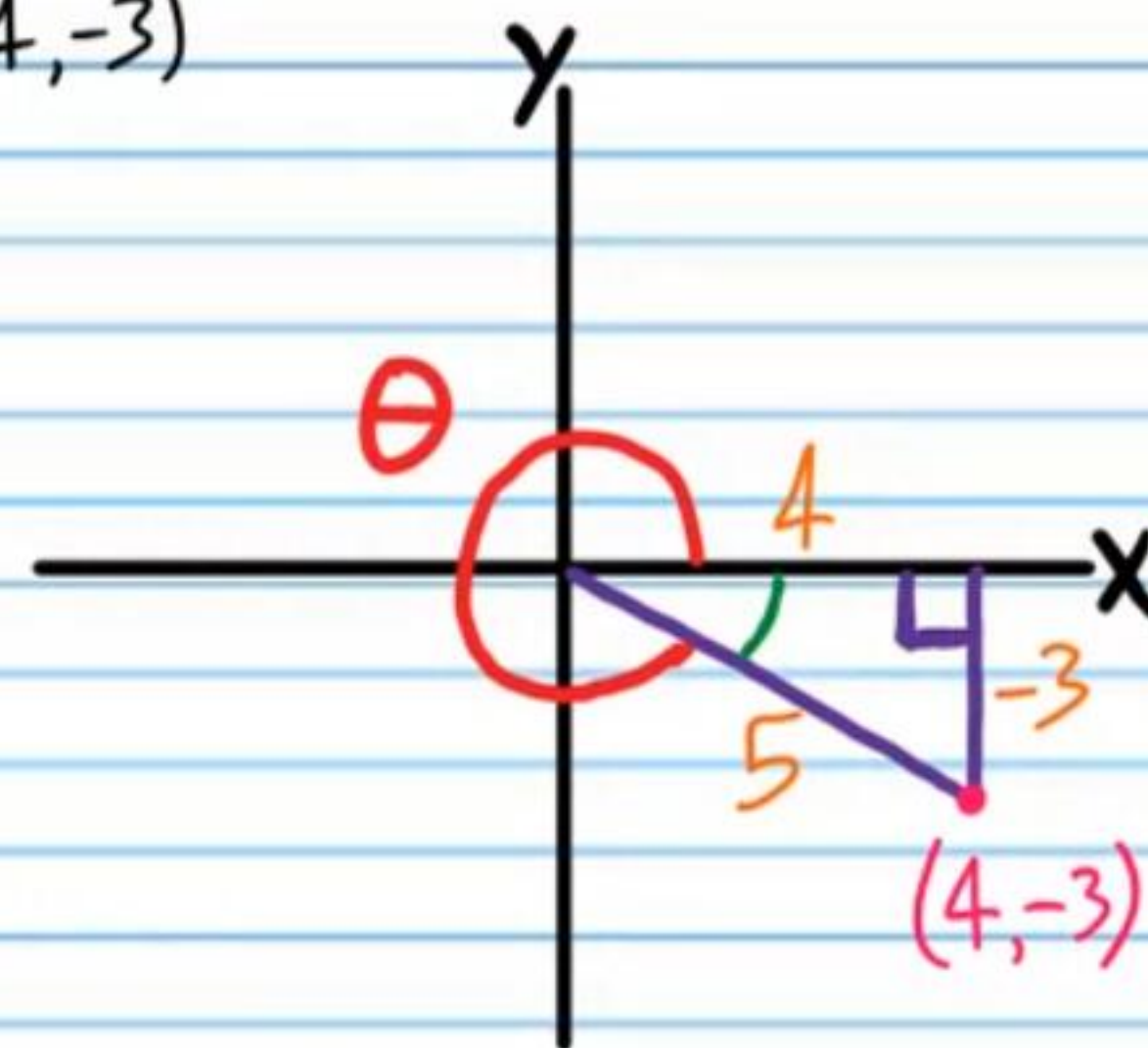
$$\text{ii) } \sec w = \boxed{\sqrt{5}}$$

$$\text{iii) } \cot w = \boxed{\frac{1}{2}}$$

$$\text{iv) } \cos w = \frac{1}{\sqrt{5}} = \boxed{\frac{\sqrt{5}}{5}}$$

$$\text{v) } \sin w = \frac{2}{\sqrt{5}} = \boxed{\frac{2\sqrt{5}}{5}}$$

(24) (4, -3)



$$\text{i) } \sin \theta = \boxed{-\frac{3}{5}}$$

$$\text{ii) } \cos \theta = \boxed{\frac{4}{5}}$$

$$\text{iii) } \tan \theta = \boxed{-\frac{3}{4}}$$

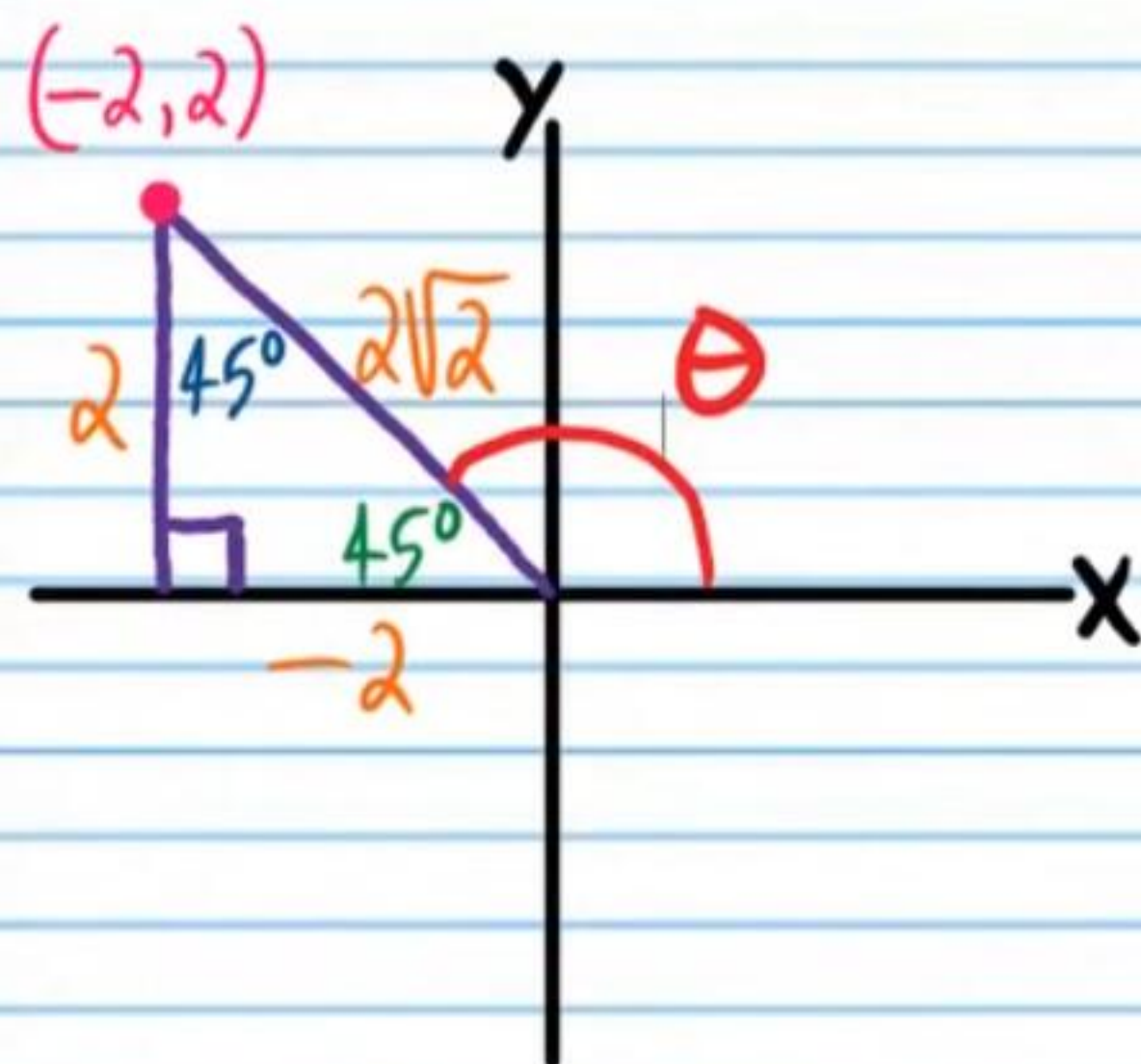
$$\text{iv) } \csc \theta = \boxed{-\frac{5}{3}}$$

$$\text{v) } \sec \theta = \boxed{\frac{5}{4}}$$

$$\text{vi) } \cot \theta = \boxed{-\frac{4}{3}}$$



25)  $(-2, 2)$



i)  $\sin \theta = \frac{1}{\sqrt{2}} = \boxed{\frac{\sqrt{2}}{2}}$

ii)  $\cos \theta = -\frac{1}{\sqrt{2}} = \boxed{-\frac{\sqrt{2}}{2}}$

iii)  $\tan \theta = \boxed{-1}$

iv)  $\csc \theta = \boxed{\sqrt{2}}$

v)  $\sec \theta = \boxed{-\sqrt{2}}$

vi)  $\cot \theta = \boxed{-1}$